

SECTION IX

MANAGEMENT OF CHRONIC KIDNEY DISEASE

Classification and Management of Chronic Kidney Disease

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CHAPTER OUTLINE

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KEY POINTS

- The 2012 Kidney Disease Improving Global Outcomes classification system for chronic kidney disease (CKD) provides simple risk stratification and serves as a convenient framework for guiding management through different stages.
- Lifestyle measures including smoking cessation, weight loss, and dietary sodium restriction should form part of any strategy to achieve kidney protection and reduce cardiovascular risk in persons with CKD.
- Evidence from The Systolic Blood Pressure Intervention Trial indicates that a lower target for the treatment of systolic blood pressure may afford improved cardiovascular risk reduction but may also be associated with some increase in risk of acute kidney injury and hyperkalemia. The risks versus benefits of a lower blood pressure target should be evaluated in each individual.
- The first pillar of treatment to achieve optimal kidney protection is treatment with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers as “first line” in persons with albuminuria.
- The second pillar of treatment for kidney protection is initiation of sodium-glucose cotransporter 2 (SGLT2) inhibitor in persons with type 2 diabetes and CKD and an eGFR of ≥ 20 mL/min/1.73 m², irrespective of UACR. Likewise, SGLT2 inhibitors should be initiated in persons without diabetes and an eGFR between 20 to 45 mL/min/1.73 m² irrespective of urine albumin-creatinine ratio (UACR) or an eGFR of >45 mL/min/1.73 m² with a UACR ≥ 200 mg/g.
- In persons with type 2 diabetes, the third and fourth pillars involve treatment with a nonsteroidal mineralocorticoid receptor antagonist and a glucagon-like peptide 1 receptor agonist for treatment for kidney and cardiovascular protection.
- To achieve optimal kidney protection, careful attention should be paid to addressing all modifiable aspects of CKD to achieve target blood pressure, minimize proteinuria, and slow glomerular filtration rate decline.
- All persons with CKD should be regarded as having a high risk of cardiovascular disease and management offered to reduce this risk, including consideration of RAAS blockade, statin, and SGLT2 inhibitor use and, in those with type 2 diabetes, a glucagon-like peptide-1 receptor agonist and finerenone.

Chronic kidney disease (CKD) is a collective term covering all primary disease processes that result in structural or functional kidney abnormalities, with implications for health, persisting for at least 3 months. Abnormal urinalysis findings identifying proteinuria or hematuria and/or abnormal kidney structure or histologic features, with or without a decreased glomerular filtration rate (GFR), are the defining manifestations.¹ The rationale for having a single term that encompasses a wide range of kidney pathologies is that damage to the kidneys results in shared pathophysiologic features and that many of the interventions to slow the progression of CKD and reduce the associated cardiovascular risk should be applied regardless of the etiology of CKD. This formal definition of CKD proposed in 2002 has had a profound impact on the practice of nephrology. Before this, it was difficult to compare the results of epidemiologic and clinical studies because each used a different definition, and imprecise terms such as chronic renal disease, impairment, insufficiency, and failure were used by different authors. Epidemiologic research based on the unified definition has revealed that CKD is common, affecting on average 13.4% of the adult population globally,^{2,4} and CKD has been identified as a leading cause of death in global burden of disease studies,⁵ though it should be noted that most epidemiologic studies rely on only a single abnormal GFR value for the diagnosis of CKD and therefore probably overestimate the true prevalence. Nevertheless, nephrology has changed from a specialty that focused largely on the treatment of rare and advanced kidney diseases, as well as kidney replacement therapy (KRT) to one that is now concerned with a condition, CKD, that affects a substantial proportion of the general population, and which may have modifiable disease trajectories.⁶ Of necessity, this has also resulted in the need for health care workers who are not nephrologists to gain knowledge of CKD and its management. Thus education of health care workers on the management of CKD has the potential to reduce the burden of disease and the need for KRT. This chapter reviews in detail the classification of CKD and the evidence-based treatments that should be offered to persons with CKD regardless of the underlying kidney pathology. Within this context, there are four broad aims:

1. Attenuate GFR decline and thus prevent or delay the need for dialysis. It is perhaps self-evident, but the earlier that CKD progression can be halted, the greater is the possibility of maintaining kidney function as close to normal as possible.
2. Prevent premature cardiovascular death at all stages of CKD.
3. Recognize and manage complications of CKD as they arise, particularly in stages 4 and 5.
4. Timely preparation for KRT or conservative and palliative care.

Because the pathophysiology associated with CKD changes as the condition progresses, we also propose a stepped care approach and consider how primary and secondary care clinicians should cooperate to provide optimal care for persons with CKD.

CLASSIFICATION OF CHRONIC KIDNEY DISEASE

The original classification proposed in 2002 subdivided CKD into five stages according to the GFR, reflecting the observation that in the majority of cases, CKD progresses slowly through the stages before reaching end-stage kidney failure (ESKF). In 2008, a National Institute for Health and Care Excellence (NICE) guideline proposed that stage 3 (GFR 59–30 mL/min/1.73 m²) be subdivided into stages 3a (GFR 59–45 mL/min/1.73 m²) and 3b (GFR 44–30 mL/min/1.73 m²), reflecting a broad consensus that there are important differences in clinical aspects and prognosis between these groups. The CKD staging system has two important implications: First, it suggests that if CKD is detected at an early stage, intervention may be possible to prevent or slow progression to more advanced stages.⁷ Second, it reflects the observation that as GFR declines, the risk profile of persons and associated complications changes. Thus the staging system provides a useful framework for structuring therapy and prioritizing interventions to produce a comprehensive strategy for the management of CKD. Although universally adopted, this classification system was limited by not reflecting the risk status of those classified, a feature that is present in many other disease classification systems. Following extensive epidemiologic investigations that identified reduced GFR and albuminuria as independent risk factors for adverse outcomes, the CKD classification system was revised by Kidney Disease Improving Global Outcomes (KDIGO) in 2012 and reaffirmed in the 2024 KDIGO guidelines for management of CKD.⁸ The GFR stages were preserved, though now termed categories G1 to G5, and new categories were added for albuminuria; categories A1 to A3, corresponding to the previously used descriptions of “normoalbuminuria,” “microalbuminuria,” and “macroalbuminuria.” In addition, it was emphasized that the etiology of CKD should be included in the classification, giving rise to the term “CGA” classification (cause, GFR, albuminuria; Fig. 54.1).^{9,10} According to this system, a person with immunoglobulin A (IgA) nephropathy, a GFR of 34 mL/min/1.73 m², and a urine albumin-to-creatinine ratio (UACR) of 367 mg/g would be classified as “IgA nephropathy, category G3b A3.” Implementation of the CGA classification has been greatly aided by the global adoption of equations to derive an estimated GFR (eGFR) from the serum creatinine concentration and use of UACR on a random sample to quantitate albuminuria (see Chapter 23). Importantly, in this classification system, each category reflects a risk status for multiple adverse outcomes including ESKF, progression of CKD, cardiovascular events (CVEs), and all-cause mortality as indicated in the accompanying heat map. Thus the category is useful to guide the therapeutic approach at each stage and prompt referral for specialist care when indicated.

Despite its global adoption, some commentators have questioned the utility of the CKD definition and classification, suggesting that the high prevalence of CKD reflects a normal aging process and pointing out that mildly reduced GFR (category G3a) in the absence of albuminuria

Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–300 mg/g 3–30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (mL/min/1.73 m ²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60–89			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	<15			

Green: low risk (if no other markers of kidney disease, no CKD)
 Yellow: moderately increased risk
 Orange: high risk
 Red: very high risk

Fig. 54.1 Current classification system and nomenclature proposed by the Kidney Disease Improving Global Outcomes (KDIGO) in 2012. Chronic kidney disease (CKD) is defined as abnormalities of kidney structure or function, present for 3 months, with implications for health. CKD is classified on the basis of cause, glomerular filtration rate (GFR) category, and albuminuria category. (With permission from Kidney Disease: Improving Global Outcomes CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int.* 2013;3(Suppl):1–150.)

is associated with minimal increase in risk, particularly in older persons.^{11,12} Various solutions have been proposed to address this issue, including the introduction of an age-adjusted GFR threshold for defining CKD or a modification of the classification for different age groups, but to date, no consensus has been reached and the KDIGO classification is used worldwide.

MECHANISMS OF DISEASE PROGRESSION AND THE RATIONALE FOR INTERVENTIONS TO ACHIEVE KIDNEY PROTECTION

It has long been appreciated that regardless of the primary kidney disease, kidney damage tends to progress toward ESKF, especially if more than half of glomeruli have been lost. This suggests that a common pathway of mechanisms may promote kidney injury and establish a vicious cycle of nephron loss. In research efforts since the 1960s, investigators have identified glomerular hemodynamic factors (glomerular hypertension and hyperfiltration), multiple effects of angiotensin II, proteinuria, and proinflammatory and profibrotic molecules as key elements of this pathway (discussed in detail in Chapter 50). A common pathway underlies progressive kidney damage from kidney diseases of diverse causes, and recognition of this pathway has been

vital in informing strategies to achieve kidney protection. Thus the interventions to slow CKD progression discussed in the following sections are each aimed at attenuating mechanisms of progression. In view of the redundancy that is characteristic of most biologic systems, it has become clear that to achieve optimal kidney protection, attempts should be made to inhibit the vicious cycle of common pathway mechanisms at multiple points (Fig. 54.2).

INTERVENTIONS FOR SLOWING PROGRESSION OF CHRONIC KIDNEY DISEASE

LIFESTYLE INTERVENTIONS

EXERCISE

All guidelines support healthy lifestyle advice and exercise for persons with CKD, citing the substantial literature that exists largely for the general population. A major issue for nephrologists and physicians managing CKD is that there has yet to be a dedicated randomized controlled trial (RCT) examining the effect of lifestyle interventions in persons with CKD. The most contemporary evidence comes from an ancillary analysis of the Lifestyle Interventions and Independence for Elders (LIFE) trial.¹³ The LIFE study

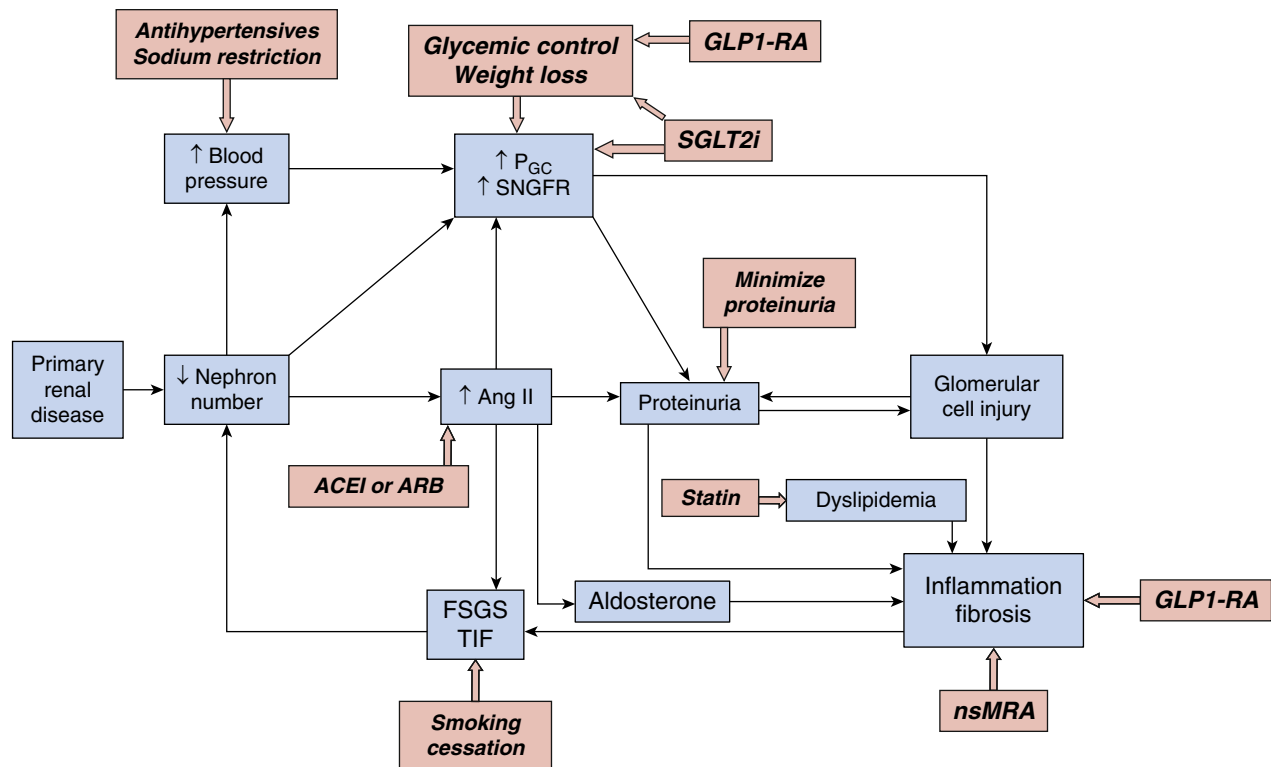


Fig. 54.2 A common pathway of mechanisms that result in a vicious circle of nephron loss in chronic kidney disease (CKD). Interventions (in red) for achieving kidney protection are directed at inhibiting the common pathway at multiple points to slow CKD progression. ACEI, Angiotensin-converting enzyme inhibitor; Ang II, angiotensin II; ARB, angiotensin receptor blocker; FSGS, focal segmental glomerulosclerosis; GLP-1 RA, glucagon-like peptide-1 receptor agonist; nsMRA, nonsteroidal mineralocorticoid antagonist; P_{GC} , glomerular capillary hydraulic pressure; SGLT2i, sodium-glucose cotransporter 2 inhibitor; SNGFR, single nephron glomerular filtration rate; TIF, tubulointerstitial fibrosis.

randomized 1635 sedentary older adults aged 70 to 89 years to a moderate-intensity physical activity and exercise program versus a “successful aging” health education program and examined clinical outcomes at 2 years. The study included 1199 participants—of whom 796 (66%) had a cystatin C–based eGFR [$eGFR_{CysC}$] <60 mL/min/1.73 m² and the outcomes of interest were the change in $eGFR_{CysC}$, as well as a rapid decline in kidney function, defined as the high tertile of change in $eGFR_{CysC}$ of $>6.7\%$ per year. Compared with the education intervention, physical activity and exercise significantly slowed the rate of $eGFR_{CysC}$ decline at 2 years (mean difference [MD], 0.96 mL/min/1.73 m²; 95% confidence interval [CI], 0.02–1.91 mL/min/1.73 m²) and lowered the odds of rapid kidney function decline by 20% (odds ratio: 0.79, 95% CI, 0.65–0.97).¹³ Subgroup analyses showed a consistent pattern of benefit in persons with and without CKD.

Beyond kidney function, evidence does exist for the value of lifestyle intervention and exercise regarding treating hypertension and preventing CVE. Therefore in the absence of dedicated clinical trials of lifestyle interventions in persons with CKD, it is reasonable to expect that a similar relative (and greater absolute) benefit might be gained in terms of CKD progression and CVE.

SMOKING CESSATION

Tobacco use is the most common cause of avoidable cardiovascular mortality worldwide; therefore not surprisingly, smoking cessation is one of the most promoted methods in management of cardiovascular risk in the

general population and in persons with CKD. Evidence that smoking cessation helps prevent CKD progression is emerging.¹⁴ In the Multiple Risk Factor Intervention Trial (MRFIT), smoking was significantly associated with an increased risk for ESKF¹⁵; in the Prevention of Renal and Vascular End-stage Disease (PREVEND) study, the rate of urine albumin excretion was correlated with the number of cigarettes smoked.¹⁶ Smoking has been identified as a risk factor for the development of microalbuminuria and overt proteinuria and for the progression of CKD in persons with type 1 and type 2 diabetes.^{17–19} In a large Swedish study, the risk for disease-specific types of CKD among smokers was compared with that among people who had never smoked. Overall, the association was modest, but an important finding was that the risk increased with high daily consumption (>20 cigarettes/day), long duration (>40 years), and a high cumulative “dose” (>30 pack-years) in comparison with participants who had never smoked. The increase in risk associated with smoking was highest for persons with CKD classified as nephrosclerosis and also glomerulonephritis.²⁰ In the Jackson Heart Study, smoking was associated with a higher risk of rapid GFR decline ($>30\%$ decline from baseline) in a dose-dependent manner²¹ and in a combined analysis of five large U.S. cohort studies ($n = 954,029$), smoking was associated with a doubling in the risk of death from ESKF.²² Finally, a meta-analysis of 15 community-based cohort studies observed an increase in the risk of developing CKD in former (hazard ratio [HR], 1.15; 95% CI, 1.08–1.23) and current smokers (HR, 1.34; 95% CI, 1.23–1.47), as well as an increased risk of ESKF in

former (HR, 1.44; 95% CI, 1.00–2.09) and current smokers (HR, 1.91; 95% CI, 1.39–2.64), though smoking was not associated with the development of proteinuria.²³

Two large cohort studies have reported detailed data on the impact of smoking on CKD progression. In the CRIC study, smoking was associated with increased all-cause mortality and the combined endpoint of all-cause mortality and CKD progression, though not with CKD progression alone.²⁴ In an analysis of data from the Study of Heart and Renal Protection (SHARP) trial, current and former smoking were associated with higher levels of proteinuria but not with CKD progression during a median follow-up of 4.9 years. Smoking was associated with significantly increased risks of CVD, cancer, and all-cause mortality.²⁵ RCTs on the effect of smoking cessation on CKD progression have yet to be published, and few prospective data are available, but in one study of persons with diabetes, smoking cessation was associated with less progression to macroalbuminuria (proteinuria) and a slower rate of GFR decline in comparison to persons who continued smoking.²⁶ In addition, the intermediate risk for adverse outcomes reported for ex-smokers versus current smokers in several of the aforementioned studies provides indirect evidence of benefit from smoking cessation. Further prospective studies are required, but published evidence strongly supports smoking cessation as an intervention to reduce the incidence and progression of CKD.

Pharmacotherapy to assist in smoking cessation is now well established. A meta-analysis of RCTs in the general population (69 trials involving a total of 32,908 persons) showed that varenicline, bupropion, and five nicotine replacement therapies were all more efficacious than placebo at promoting smoking abstinence at 6 and 12 months. In comparison with persons taking placebo, persons treated with these agents were 1.5 to 2.5 times more likely to quit smoking, depending on the specific agent.²⁷ Indeed, combining varenicline and a nicotine replacement patch achieved an impressive higher continuous abstinence rate at 12 weeks (55.4% vs. 40.9%) and 6-month point prevalence abstinence rate of 65.1% versus 46.7% compared with varenicline treatment alone.²⁸ An important finding is that people who smoke are more likely to stop smoking if offered a combination of interventions, such as behavioral support and pharmacotherapy. Multicomponent interventions are now part of many public health guidelines. We see no reason to exclude people with CKD from this advice.

WEIGHT LOSS

Obesity is the dominant risk factor for type 2 diabetes and is also a major risk factor for hypertension and progression of CKD. Evidence linking the metabolic syndrome and CKD has emerged; each element of the metabolic syndrome is associated with increased prevalence of CKD and microalbuminuria. In animal models (obese Zucker rats with type 2 diabetes), early progressive podocyte damage and macrophage infiltration are associated with hyperlipidemia and antedate both the development of glomerulosclerosis and tubulointerstitial damage.^{29,30} In humans, there was a graded relationship between the number of components of the metabolic syndrome present and the corresponding prevalence of CKD or microalbuminuria.³¹

Epidemiologic studies have identified obesity as a risk factor for CKD,^{32,33} and in one study, obesity was an independent risk factor for progression of IgA nephropathy.³⁴ Furthermore, the largest such study to assess risk of CKD in association with obesity demonstrated a strong biologic gradient: Increasing body mass index (BMI) was associated with increasing risk of ESKF. In comparison with persons with an “ideal” BMI (18.5–24.9 kg/m²), the relative risk (RR) of ESKF was 3.6-fold for those with a BMI of 30 to 34.9 kg/m², sixfold for those with BMI of 35 to 39.9 kg/m², and sevenfold for those with BMI of 40 kg/m² or higher. Controlling for baseline blood pressure and presence of diabetes attenuated the associations, but the gradient between increasing body size and ESKF risk remained strong.³⁵

Because these components are major factors in the initiation and progression of CKD, respectively, there has been significant focus on the potential for benefit from weight loss to reverse the features of the metabolic syndrome, as observed in the general population.³⁶ Results of early studies support this assertion, inasmuch as weight loss in humans with obesity demonstrated reversal of glomerular hyperfiltration and albuminuria.³⁷

Moreover, there is evidence that weight loss of as little as 10 lb (4.5 kg) reduces blood pressure, prevents hypertension, or does both in a large proportion of overweight persons, although the ideal is to maintain normal body weight.³⁸ In the Framingham Heart Study, weight loss of 5 lb (2.25 kg) or more was associated with reductions in cardiovascular risk of about 40%³⁹ for both men and women and thus should be a clear goal for persons with CKD who are overweight. It also appears that the degree of weight loss, regardless of method (lifestyle changes or bariatric surgery), dictates the benefits of lowering blood pressure and reduction in glycemic markers.⁴⁰ However, according to longer-term studies of lifestyle modification and of persons after bariatric surgery, blood pressure lowering benefits regress somewhat over time, although the vascular outcomes continue to be better than those in control groups.^{36,41,42}

Kidney-protective effects associated with weight loss interventions (dietary caloric restriction, exercise, antiobesity medications, and bariatric surgery) were reported in a meta-analysis of data from 522 participants in five controlled and eight uncontrolled trials.⁴³ In persons undergoing interventions to lose weight, those with proteinuria had a mean reduction of urine protein levels of 1.7 g/day; even among persons with microalbuminuria, mean reduction of urinary albumin excretion was 14 mg/day. Although these reductions were modest in comparison with those in persons with overt proteinuria, results of other studies of blood pressure and glycemic control interventions on microalbuminuria suggest that they will have long-term benefit in such persons.⁴³ This was confirmed by a secondary analysis of the Look AHEAD trial in which 5145 overweight or obese persons with type 2 diabetes were randomly assigned to an intensive lifestyle intervention (ILI) or diabetes support and education (DSE). The trial was stopped early due to lack of any difference in the primary outcome, CVDs, but analysis of kidney outcomes showed a 31% reduction in the incidence of “high-risk CKD” as defined by the KDIGO classification system (HR, 0.69; 95% CI, 0.55–0.87) in the

ILI versus DSE group, mediated in part by weight loss and improvements in hypertension and glycemic control.^{44, 45}

With respect to weight management in people with CKD, glucagon-like peptide 1 receptor agonists (GLP1RAs) have emerged as important pharmacotherapeutic options. GLP1RAs are long-acting analogs of the naturally occurring incretin hormone, glucagon-like peptide-1 (GLP1), which is responsible for glucose-dependent insulin secretion. Initially developed as glucose-lowering medications, GLP1RAs also has notable cardiometabolic benefits including weight loss. The effect of GLP1RAs on weight loss is mediated through its effect on GLP1 receptors in multiple brain regions ultimately reducing appetite and caloric intake and increasing satiety.⁴⁶ Compared with lifestyle modification in people with type 2 diabetes, GLP1RAs reduce body weight by an average of 5.79 kg (MD -5.79, 95% CI: -6.3 kg to -5.3 kg), with particularly sizable weight loss with semaglutide (MD -11.40 kg, 95% CI: -12.5 kg to -10.3 kg).⁴⁷ These benefits also extend to people with moderate kidney impairment,^{48,49} without any evidence of a disproportionate increase in adverse events. Indeed, currently available GLP1RA formulations such as semaglutide, liraglutide, and dulaglutide have minimal renal excretion and do not require dose adjustment.⁵⁰⁻⁵⁴ These formulations have also been used off-label in people with ESKF, although dedicated studies on their safety and efficacy have yet to be conducted.

In addition to GLP-1, glucose-dependent insulinotropic polypeptide (GIP) is the only other known incretin hormone. In dedicated RCTs focused on weight management, tirzepatide, a coagonist of the GLP1 and GIP receptor, resulted in greater weight loss than the selective GLP1RA, semaglutide. A more detailed discussion of the kidney-specific effects of GLP1RAs and GIP receptor agonists is provided later in “Glucagon-Like Peptide-1 Receptor Agonists and Glucose-Dependent Insulinotropic Polypeptide Receptor Agonists.”

Bariatric surgery is an important adjunctive strategy for weight management, and its increased use in people with CKD has provided an opportunity to study its effect on kidney function. In one long-term study of 2144 persons who underwent bariatric surgery, kidney outcomes were assessed after 7 years on the basis of the change in CKD risk category as defined by the KDIGO classification. Among those with moderate risk at baseline, the risk category improved in 53% and deteriorated in 5% to 8%; in the high-risk group, improvement was observed in 56% and deterioration in 3% to 10%; in the very-high-risk group, 23% improved. eGFR initially improved with a peak at 2 years and then gradually declined. Albuminuria showed large and sustained decreases in moderate (median UACR 48–14 mg/g) and high-risk groups (median UACR 326–26 mg/g).⁵⁵ In another study that included only persons with CKD stages 3 and 4 and propensity score matching with similarly obese persons who did not undergo bariatric surgery, eGFR was significantly higher by a mean of 9.84 mL/min/1.73 m² in the surgery group after 3 years.⁵⁶ These studies are limited by relying on eGFR to assess kidney function, but the finding that benefit was sustained over several years provides reassurance that the improvement was not simply an artifact due to early decrease in serum creatinine secondary to weight loss after surgery. In a meta-analysis that included 23 cohort studies and 3015 persons who underwent all forms of bariatric

surgery, small but statistically significant improvements were observed in serum creatinine (mean decrease 0.08 mg/dL) and proteinuria (mean decrease 0.04 g/day).⁵⁷ Finally, the Microvascular Outcomes after Metabolic Surgery (MOMS) RCT compared the effect of Roux-en-Y gastric bypass (RYGB) surgery versus best medical treatment in 100 participants with early stage CKD (eGFR ≥ 30 mL/min/1.73 m² and UACR ≥ 30 mg/g), type 2 diabetes, and a BMI of 30 to 35 kg/m². RYGB surgery resulted in a 28% increase in the percentage of participants with albuminuria remission at 2 years compared with medical treatment.⁵⁸

On the basis of available data, we recommend weight loss in obese persons with CKD through a combination of increased exercise, reduced caloric intake, pharmacologic therapies, and consideration of bariatric surgery if these are not successful.

DIETARY INTERVENTIONS

DIETARY SODIUM RESTRICTION

Salt (sodium chloride) intake in many countries is 9 to 12 g/day. However, the World Health Organization and KDIGO recommendation for adults is to reduce salt intake to 5 g/day or less on the basis of substantial evidence from epidemiologic, migration, intervention, genetic, and animal studies that salt intake plays an important role in regulating blood pressure.⁸

Essential hypertension is observed primarily in societies in which the average sodium intake exceeds 100 mEq/day (2.3 g sodium or ≈ 6 g sodium chloride) and is rare in societies in which the average sodium intake is less than 50 mEq/day (1.2 g sodium or 3 g sodium chloride).⁵⁹ Furthermore, in a subanalysis of the Prospective Urban Rural Epidemiology (PURE) study—a prospective cohort study of 157,543 adults 35 to 70 years of age from 18 low-, middle-, and high-income countries—the association between sodium excretion and blood pressure was also nonlinear. Among all participants included in the PURE, each 1 g increase in estimated 24-hour sodium excretion was associated with a 2.11 mm Hg increase in systolic blood pressure and 0.78 mm Hg increase in diastolic blood pressure.⁶⁰ The magnitude of this association was larger with higher sodium intake. Among persons with an estimated 24-hour sodium excretion > 5 g per day, the increment in systolic blood pressure per 1 g change in sodium excretion was 2.58 mm Hg, compared with 1.74 mm Hg per gram of sodium excretion for 3 to 5 g per day, and 0.74 mm Hg per g of sodium excretion for < 3 g per day ($P < 0.001$ for interaction). The magnitude of the association between each gram of estimated 24-hour sodium excretion and systolic blood pressure was also larger for persons with hypertension than for those without hypertension and with increased age.

Accordingly, sodium restriction produces a significant reduction in blood pressure.^{61,62} A meta-analysis of RCTs with a duration of at least 4 weeks concluded that reducing salt intake by 3 g/day is predictive of a linear fall in blood pressure of, on average, 3.6 to 5.6 mm Hg (systolic blood pressure [SBP]) and 1.9 to 3.2 mm Hg (diastolic blood pressure [DBP]) in hypertensive persons and of 1.8 to 3.5 mm Hg (SBP) and 0.8 to 1.8 mm Hg (DBP) in normotensive persons.⁶³ Weight loss and reduced sodium intake are particularly beneficial in older people. In the

Trial of Nonpharmacologic Interventions in the Elderly (TONE), reducing sodium intake to 80 mEq (2 g) per day reduced blood pressure over 30 months, and about 40% of participants on the low-salt diet were able to discontinue their antihypertensive medications.⁶⁴

In many persons with CKD, especially those with glomerular disease and severe proteinuria, hypertension behaves in a salt-sensitive manner. In one large observational study of persons without CKD at baseline, both high (>4.03 g/day of sodium) and low (<2.08 g/day of sodium) dietary sodium intake was associated with an increased incidence of CKD during follow-up of more than 10 years in 3106 hypertensive participants, but no association with sodium intake was observed in those without hypertension ($n = 4871$).⁶⁵ This supports that salt sensitivity is important in the association between sodium intake and CKD. The reason for the association with low sodium intake is unknown but may be confounded by associations between sodium intake and several other nutritional factors including potassium, total calorie, fat, protein, and carbohydrate intake, though the analysis was adjusted for these factors. A systematic review of 16 small studies that investigated the association of dietary sodium intake and CKD progression concluded that marked heterogeneity between the studies precluded meta-analysis.⁶⁶ Nevertheless, the general trend observed was that increasing sodium intake is associated with worsening albuminuria. An analysis of 3757 participants with CKD from the CRIC study found that after 15,807 person-years of follow-up, the highest quartile of urinary sodium excretion (≥ 195 mmol/day) was associated with a significantly increased risk of CKD progression (HR, 1.54; 95% CI, 1.23–1.92) and all-cause mortality (HR, 1.45; 95% CI, 1.08–1.95) compared with the lowest quartile (<117 mmol/day).⁶⁷

High dietary sodium intake has also been shown to negate the antiproteinuric effects of treatment with angiotensin-converting enzyme (ACE) inhibitors.⁶⁸ In a prospective, randomized, placebo-controlled crossover study, dietary sodium restriction increased the antihypertensive and antiproteinuric effects of therapy with angiotensin receptor blockers (ARBs), as monotherapy or in combination with a thiazide diuretic, in persons with nondiabetic CKD (Fig. 54.3).⁶⁹ Whereas the protocol aimed to restrict dietary salt intake to less than 50 mEq/day (1.2 g of sodium or 3 g of sodium chloride), these impressive results were observed with achieved sodium restriction of only 92 mEq/day.⁶⁹ Furthermore, a post hoc analysis of data from the first and second Ramipril Efficacy in Nephropathy (REIN) trials found that medium and high sodium intake were associated with significant increases in the incidence of ESKF versus low sodium intake. Each 100-mEq/g increase in 24-hour urinary sodium/creatinine excretion was associated with a 1.61-fold (95% CI, 1.15–2.24) higher risk of ESKF, independent of blood pressure.⁷⁰

Several small prospective studies have investigated the impact of dietary sodium restriction on blood pressure and proteinuria in persons with CKD. In a randomized placebo-controlled crossover study (achieved using low sodium diet of 60 to 80 mmol/day plus sodium chloride tablets [120 mmol/day] or placebo) in 20 people with stage 3 or stage 4 CKD, low sodium intake was associated with an average 10/4 mm Hg reduction in blood pressure, as well as reductions

in albuminuria and proteinuria.⁷¹ In a similar randomized crossover study, dietary sodium restriction to less than 2 g/day for 4 weeks in persons with CKD stage 3 and 4 was associated with an average reduction in extracellular volume of 1.02 L, weight loss of 2.3 kg, and mean reduction in ambulatory SBP of 10.8 mm Hg but no improvement in albuminuria.⁷² By contrast, another study in persons with CKD stage 1–3 and persistent albuminuria greater than 300 mg/day despite treatment with ramipril 10 mg/day reported that dietary sodium restriction for 8 weeks was associated with a reduction in arterial pressure from a mean of 95 to 90 mm Hg and a significant decrease in albuminuria from a mean of 1060 to 717 mg/day. Creatinine clearance also decreased from a mean of 101 to 91 mL/min.⁷³ Another study illustrated the challenges associated with achieving sustained reduction in sodium intake. In an RCT, persons with CKD assigned to a 3-month intervention of education and coaching, as well as self-monitoring, achieved a modest mean reduction in urinary sodium excretion of 30 mmol/day associated with a mean reduction in ambulatory DBP of 3.4 mm Hg and mean reduction in proteinuria of 0.4 g/day, but at 3 months after completion of the intervention there was no difference in urinary sodium excretion or ambulatory BP compared with controls, although proteinuria remained 0.3 g/day lower.⁷⁴ Further long-term randomized trials are needed to define the role of sodium restriction in kidney-protective strategies, but even the incomplete evidence available supports a recommendation for moderate dietary sodium restriction to less than 5 g/day of salt in persons with CKD.

Food processing drastically changes the cationic content of natural foods, increasing sodium content and decreasing potassium content. On average, approximately 10% of dietary sodium chloride originates naturally in foods, whereas approximately 80% is the result of food processing, the remainder being discretionary (added during cooking or at the table). We advocate assessment of salt intake in individuals with CKD and advise reducing salt intake with the assistance of a dietitian to less than 5 g/day (<90 mmol/day of sodium) as recommended by KDIGO.⁸

DIETARY PROTEIN RESTRICTION

Dietary protein restriction as a kidney-protective strategy is based on the notion that reducing the excretory burden on the kidneys would slow the rate of progressive injury. Unfortunately, clinical studies have failed to provide clear evidence to support the use of protein restriction in human CKD.

The Modification of Diet in Renal Disease (MDRD) study was designed to provide a definite answer to the question of whether dietary protein restriction slows the progression of CKD. The study had two components: In study A, 585 persons with mostly nondiabetic CKD (GFR, 25–55 mL/min/1.73 m²) were randomly assigned to follow either a diet with “usual” protein levels (1.3 g/kg/day) or a low-protein diet (LPD) (0.58 g/kg/day); in study B, 255 persons with GFRs of 13 to 24 mL/min/1.73 m² were randomly assigned to follow either LPD (0.58 g/kg/day) or “very low” protein diet (VLPD; 0.28 g/kg/day) with keto-amino acid supplementation to prevent malnutrition. After a mean follow-up period of 2.2 years, there was no difference in the rate of GFR decline in study A and only a

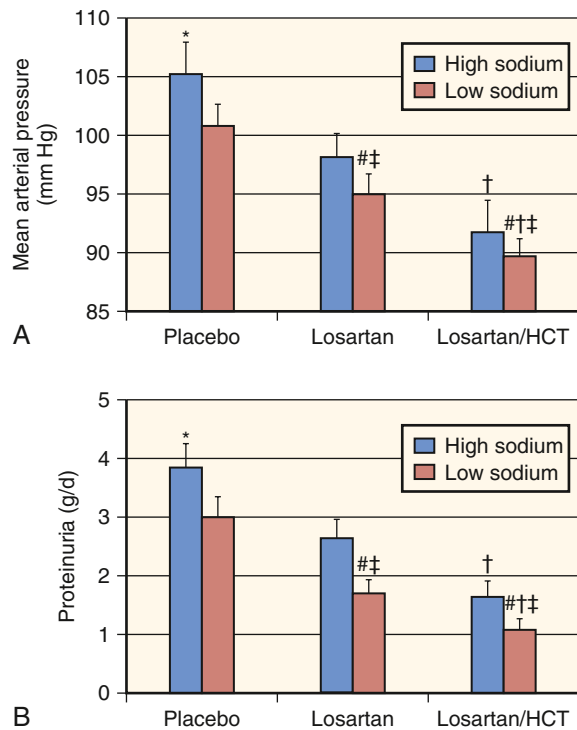


Fig. 54.3 Results of a prospective randomized crossover trial showing the effect of dietary sodium restriction on the (A) antihypertensive and (B) antiproteinuric effects of treatment with an angiotensin receptor blocker as monotherapy or in combination with a thiazide diuretic. * $P < .05$ versus all periods; # $P < .05$ versus same treatment on high-salt diet (effect of low-salt diet); † $P < .05$ versus losartan treatment on same diet [effect of hydrochlorothiazide (HCT)]; ‡ $P < .05$ versus placebo on same diet. (From Vogt L, Waanders F, Boomsma F, et al. Effects of dietary sodium and hydrochlorothiazide on the antiproteinuric efficacy of losartan. *J Am Soc Nephrol*. 2008;19:999–1007.)

trend toward slower decline in the VLPD group in study B.⁷⁵ Further analysis indicated, however, that the desired protein intake was not achieved in the randomized groups, and secondary analysis based on achieved dietary protein intake demonstrated that a reduction in protein intake of 0.2 g/kg/day was correlated with a 1.15-mL/min/year reduction in the rate of GFR decline, equivalent to a 29% reduction in mean rate of GFR decline.⁷⁶ In addition, a post hoc two-slope analysis, in which presumptive acute effects of dietary protein restriction were taken into account, suggested a modest long-term benefit. However, long-term follow-up of the cohort in MDRD study A yielded disappointingly inconclusive results.⁷⁷ In a more recent trial, 207 well-nourished persons without diabetes and with eGFR less than 30 mL/min/1.73 m² and urine protein-to-creatinine ratio (PCR) less than 1 g/g were randomized to a vegetarian VLPD (<0.3 g/kg/day) with ketoanalog supplementation (ketoanalog diet [KD]) or standard (nonvegetarian) LPD (<0.6 g/kg/day). After 15 months the primary outcome of RRT or 50% reduction in eGFR was reached in 13% of those on KD versus 42% on LPD (adjusted HR, 0.1; 95% CI, 0.05–0.2). Those on the KD also evidenced a 3.2 mL/min/year slower rate of eGFR decline. Metabolic factors were also improved by the KD; serum bicarbonate and calcium were higher and phosphate was lower. There was no change in

nutritional status in either group and no adverse reactions were reported.⁷⁸ However, it is unclear what proportion of the benefit was attributable to the vegetarian VLPD versus the ketoanalog supplementation.

Fouque and Laville⁷⁹ performed a Cochrane Database systematic review of all randomized studies, comparing two different levels of protein intake in adult persons suffering from moderate to severe CKD. A total of 2000 nondiabetic persons were identified in 10 studies (of a total of 40 studies) in which follow-up lasted at least 1 year. There were 281 renal deaths (progression to ESKF) recorded, 113 among participants following the LPD and 168 among those following the higher-protein diet (RR, 0.68; 95% CI, 0.55–0.84; $P = .0002$). The authors concluded that reducing protein intake in persons with CKD reduces the occurrence of renal death by 32% in comparison with higher or unrestricted protein intake. The most recent meta-analysis included 16 RCTs of persons with relatively advanced CKD (stage 4 and 5 in most cases) not receiving RRT. Inclusion required at least 30 participants per study. In studies comparing LPD (<0.8 g/day) with higher-protein intake, LPD was associated with a small but significant absolute risk reduction for ESKF (4%) and higher serum bicarbonate at 1 year (weighted mean difference [MD], 1.46 mEq/L) versus higher-protein diets. Similarly, in studies comparing VLPD with LPD, VLPD was associated with an absolute risk reduction for ESKF (13%) and higher GFR at 1 year (weighted MD, 3.95 mL/min/1.73 m²) versus LPD. No studies reported increased risk of protein-energy wasting or other safety concerns associated with dietary protein restriction.⁸⁰

The caveats for these trials are that most were of short duration, the largest trial (MDRD) mostly excluded persons with diabetes, and compliance with LPDs was a factor in interpretation of some of the results. Furthermore, the proportion of participants treated with renin-angiotensin-aldosterone system inhibitors (RAASi) was variable and the trials were conducted before the availability of SGLT2 inhibitors. Available evidence suggests that LPD and RAASi may have synergistic kidney-protective effects, but this has not yet been adequately evaluated in an RCT.⁸¹ In addition to possible kidney-protective effects, dietary protein restriction results in reduced intake of sodium, phosphate, and acid, all of which may be beneficial in CKD.⁸²

In summary, published data indicate that there is likely some kidney-protective benefit from dietary protein restriction but conclusive evidence is lacking, in particular for additional benefit in persons already treated with RAASi and or SGLT2 inhibitors. The benefits are more apparent in persons with proteinuria and lower GFR, and the risks are greater in those at risk of malnutrition, such as elderly persons (age >75 years); those with below-average BMI (<20 kg/m²), muscle wasting, or myopathic symptoms; and those with evidence of protein-energy wasting. We therefore recommend that the potential benefits and risks should be considered carefully on an individual basis and with the assistance of a trained kidney dietitian. The 2024 KDIGO guideline for the management of CKD recommends avoidance of a high-protein diet and maintaining dietary protein intake at 0.8 g/kg/day in adults with CKD.⁸ For further discussion of dietary aspects in kidney disease, see [Chapter 55](#).

GLYCEMIC CONTROL IN PERSONS WITH DIABETES

The role of glycemic control in protecting the kidneys in persons with diabetes is discussed fully in [Chapter 41](#). In summary, the benefit of tight glycemic control in ameliorating diabetic nephropathy seems to decrease as CKD progresses; the greatest benefits are observed in stages 1 and 2. In addition to the kidney-protective effects, however, there is clear evidence that improving glycemic control can reduce the risk of developing other microvascular and macrovascular complications such as blindness and cardiovascular disease.⁸³ Of importance is that these benefits were maintained for up to 30 years after the trial, even though intensive control regressed to standard control levels.^{84,85} Achieving optimal glycemic control should therefore be an important goal for all persons with diabetes and CKD but should be balanced against the risk of developing hypoglycemia, which may be increased in the elderly and those with more advanced CKD. The KDIGO guidelines recommend a target hemoglobin A_{1c} of 7.0% (53 mmol/mol) to prevent or delay progression of microvascular complications, including diabetic kidney disease, unless persons have multiple comorbidities, have reduced life expectancy, or are at risk of hypoglycemia.⁹

Despite the importance of glycemic control, new classes of glucose-lowering medications such as SGLT2 inhibitors and GLP1RAs have important beneficial effects on albuminuria, CKD progression, and CV events that are independent of glycemic control and HbA_{1c}. The mechanism of action and the use of these medications for kidney protection are discussed more fully in subsequent sections (see “[Sodium-Glucose Cotransporter 2 \(SGLT2\) Inhibitors](#)” and “[Glucagon-Like Peptide-1 Receptor Agonists and Glucose-Dependent Insulinotropic Polypeptide Receptor Agonists](#)”).

TREATMENT OF HYPERTENSION

The treatment of systemic hypertension was the first intervention shown to significantly slow the rate of CKD progression, and it remains a fundamental kidney-protective strategy. Mogensen⁸⁶ and Parving and colleagues⁸⁷ pioneered the role of blood pressure control in studies of persons with type 1 diabetes, in whom the initiation of antihypertensive therapy significantly slowed the rate of GFR decline.

Similar observations were subsequently reported among persons with nondiabetic forms of CKD.⁸⁸⁻⁹⁰ In one of the earliest meta-analyses in nephrology, Kasiske and colleagues⁹¹ studied 100 controlled and uncontrolled studies that provided data on kidney function, proteinuria, or both before and after treatment with antihypertensive agents in people with diabetes. Reductions in proteinuria from antihypertensive agents that do not inhibit the renin-angiotensin-aldosterone system (RAAS) could be attributed entirely to changes in blood pressure. Furthermore, blood pressure reduction was associated with a relative increase in GFR (3.70 ± 0.92 mL/min for each 10-mm Hg reduction in mean arterial pressure [MAP]; $P = .0002$).

A large body of evidence supports the use of ACE inhibitors or ARB agents (RAAS inhibitors) as first-line antihypertensive therapy in persons with CKD (see the “[Pharmacologic Inhibition of the Renin-Angiotensin-Aldosterone System](#)”

section under “[Interventions for Slowing Progression of Chronic Kidney Disease](#)”). In the meta-analysis by Kasiske and colleagues, multiple linear regression analysis indicated that ACE inhibitors decreased proteinuria independently of changes in blood pressure, treatment duration, type of diabetes or stage of nephropathy, and study design.⁹¹ In comparison with other agents, ACE inhibitors had an additional favorable effect on GFR that was independent of blood pressure changes (3.41 ± 1.71 mL/min; $P = 0.05$).

Despite the evidence supporting the use of RAAS inhibitors, data from the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT) have been misinterpreted by some as implying that the choice of antihypertensive drug does not affect kidney outcomes in persons with CKD. Note that ALLHAT was designed to investigate the effect of antihypertensive drugs on cardiovascular rather than kidney outcomes in persons who had hypertension and at least one cardiovascular risk factor. There was no significant difference in the incidence of the primary outcome of fatal or nonfatal myocardial infarction among persons randomly assigned to receive treatment with a thiazide diuretic, a calcium channel blocker, or an ACE inhibitor.⁹² Whereas a post hoc analysis also showed no difference in the secondary outcome of ESKF or more than a 50% decrease in GFR, persons with serum creatinine levels exceeding 2 mg/dL (>170 μ mol/L) were specifically excluded from the study, and therefore only a minority (5662 of 33,357) had CKD, mostly in stages 1 to 3. Furthermore, the presence of proteinuria was not assessed.⁹³

In addition to RAAS inhibitor therapy, thiazide and other diuretics are valuable and sometimes essential as additional antihypertensive agents to achieve optimal blood pressure control. Studies have shown that high dietary sodium intake may abrogate the antiproteinuric effect of ACE inhibitor treatment, but addition of a thiazide diuretic restores the antiproteinuric effect despite ongoing high sodium intake.⁶⁸ Similarly, addition of a thiazide diuretic to ARB treatment reduced blood pressure and proteinuria in persons with IgA nephropathy.⁹⁴ We therefore recommend a thiazide diuretic as second-line antihypertensive therapy in persons who have not achieved adequate blood pressure control with an ACE inhibitor or an ARB alone, and thiazides retain their efficacy even in people with stage 4 kidney disease.⁹⁵

TRIALS OF LOW VERSUS USUAL TARGETS FOR LOWERING BLOOD PRESSURE

Despite the importance of blood pressure control in achieving kidney protection, the optimal blood pressure target remains uncertain, particularly in elderly persons and those with mild proteinuria. In several RCTs, researchers have investigated whether blood pressure targets lower than previously recommended levels afford greater kidney protection than “usual” blood pressure control, but these studies have not provided a conclusive answer. In the MDRD study, the primary analysis showed no significant difference in the rate of GFR decline between persons randomly assigned to a low blood pressure target (MAP of <92 mm Hg [equivalent to $<125/75$ mm Hg] if 18 to 60 years or MAP <98 mm Hg or persons aged 61 years or older) versus a lower blood pressure target (MAP <107 mm Hg [equivalent to $140/90$ mm Hg] or MAP <13 mm Hg, respectively). Persons assigned to achieve lower blood pressure, however,

evidenced an early rapid decrease in GFR, probably related to kidney hemodynamic effects, that obscured a later slower rate of GFR decline. Furthermore, a secondary analysis did show benefits associated with the lower blood pressure target among persons with more severe baseline proteinuria (urine protein level >1 g/day). Further secondary analysis revealed that lower achieved blood pressure was also associated with a slower GFR decline, an effect that was more marked among persons with more severe baseline proteinuria.⁷⁵ The authors concluded by recommending a blood pressure goal of $<125/75$ mm Hg (MAP, 92 mm Hg) for persons with CKD whose urine protein levels exceed 1 g/day and a goal of $<130/80$ mm Hg (MAP, 98 mm Hg) for those with urine protein levels of 0.25 to 1.0 g/day.⁹⁶

Findings of prolonged follow-up of the participants in the MDRD study suggest that the benefits of lower blood pressure may become evident only over a longer period. Analysis after almost 10 years revealed a significant reduction in the risk of ESKF (adjusted HR, 0.68) and a combined endpoint of ESKF or death (adjusted HR, 0.77) among persons randomly assigned to achieve the lower blood pressure targets. An important caveat is that treatment and blood pressure data were not available beyond the 2.2 years of the original trial.⁹⁷ In the African American Study of Kidney Disease (AASK), no significant difference in the rate of GFR decline was observed among participants randomly assigned to achieve a MAP goal of 92 mm Hg or lower versus a goal of 102 to 107 mm Hg. One possible explanation for this outcome is that participants in the AASK generally had lower baseline proteinuria (mean urine protein excretion, 0.38–0.63 g/day).⁹⁸ The results are therefore consistent with those of the MDRD study, which showed benefit only in persons with significant proteinuria. Likewise, in the REIN-2 trial, additional blood pressure reduction (blood pressure of $<130/80$ mm Hg vs. DBP of <90 mm Hg, regardless of SBP) failed to provide additional kidney protection in persons with nondiabetic CKD who were already receiving ACE inhibitor treatment.⁹⁹ Possible explanations are that the degree of additional blood pressure reduction was modest (4.1/2.8 mm Hg) and that in the group undergoing intensive blood pressure reduction, the number of participants with moderate to heavy proteinuria was small.

The Systolic Blood Pressure Intervention Trial (SPRINT) is the largest randomized trial to date to compare the effect of different BP targets on cardiovascular and kidney outcomes in persons with CKD. The study included 9361 participants aged 50 years and over with BP 130 to 180 mm Hg and increased cardiovascular risk (including all with CKD) who were randomly allocated to a systolic BP target of <120 mm Hg or <140 mm Hg. Persons with diabetes, proteinuria of >1 g/day, adult polycystic kidney disease, previous stroke, and cardiac failure were excluded. The main trial was stopped early due to evidence of significant benefit in the primary outcome of CVE and all-cause mortality in the lower BP group.¹⁰⁰ A prespecified subgroup analysis showed no evidence of effect modification by CKD status on the primary outcome of CVE or all-cause mortality. In the CKD subgroup, participants in the lower BP group evidenced a lower incidence of CVE (HR, 0.81; 95% CI, 0.63–1.05) and all-cause mortality (HR, 0.72; 95% CI, 0.53–0.99; Fig. 54.4). Importantly, these benefits were also observed in participants aged 75 years and older. There

was no difference between treatment groups in the kidney outcome of 50% decline in GFR or ESKF after a median of 3.3 years. The lower BP target was associated with some initial decline in eGFR, likely due to increased use of RAASi, whereas a small initial increase in GFR was observed with the higher BP. Nevertheless, after excluding the first 6 months of observation, the low BP target was associated with a slightly higher rate of GFR decline (0.47 mL/min/year vs. 0.32 mL/min/year; $P < 0.03$). The lower BP target was associated with lower UACR at all time points to 48 months. There was no difference in all serious adverse events between the groups. However, the low BP target was associated with a higher incidence of hyperkalemia, hypokalemia, and AKI¹⁰¹ (predominantly stage 1 and 2 and most had recovery of kidney function).¹⁰² In summary, SPRINT found that a lower BP target than previously recommended was associated with a significant reduction in CVE and all-cause mortality. There was no benefit in kidney outcomes but also no evidence of harm (apart from a small increase in incidence of AKI). It should be remembered that persons with advanced or severe CKD (evidenced by proteinuria >1 g/day) and those with diabetes were excluded. The findings are therefore consistent with those of the MDRD and AASK studies, confirming that there is little medium-term kidney-protective benefit associated with low BP targets in mild and nonproteinuric CKD. However, the results do indicate significant survival and cardiovascular benefits with lower BP targets and suggest that the benefit of lower BP targets outweighs the risks in older people.

Because not all the persons in the aforementioned trials received ACE inhibitor treatment, it remained unclear how important the level of blood pressure attained was in persons with CKD who were receiving a RAAS inhibitor. Several studies have sought to address this issue. Among persons with type 1 diabetes and established nephropathy who were receiving ACE inhibitor treatment, attainment of a “low” (MAP, 92 mm Hg) versus “usual” (MAP, 100–107 mm Hg) target blood pressure was associated with significantly milder degrees of proteinuria after 2 years, but there was no significant difference in GFR.¹⁰³ The Effect of Strict Blood Pressure Control and ACE Inhibition on the Progression of Chronic Renal Failure in Pediatric Patients (ESCAPE) trial was conducted to investigate the role of blood pressure control in children with CKD who were receiving treatment with an ACE inhibitor.¹⁰⁴ This is an important trial inasmuch as the primary endpoint (time to 50% decline in GFR or progression to ESKF) was not as affected by the competing mortality effects that complicate studies in older persons. A total of 30% of the participants who received intensified treatment for blood pressure control reached the primary endpoint, in comparison with 42% of those who received conventional treatment for blood pressure control (HR, 0.65; 95% CI, 0.44–0.94; $P = 0.02$). Urine protein excretion gradually rebounded during ongoing ACE inhibition after an initial 50% decrease, despite persistently good blood pressure control. Achievement of blood pressure targets and a decrease in proteinuria were significant independent predictors of delayed progression of CKD.

Results of secondary analyses of other studies also indicate that lower blood pressure is associated with more effective kidney protection in persons receiving RAAS inhibitor treatment. In the Irbesartan in Diabetic Nephropathy Trial

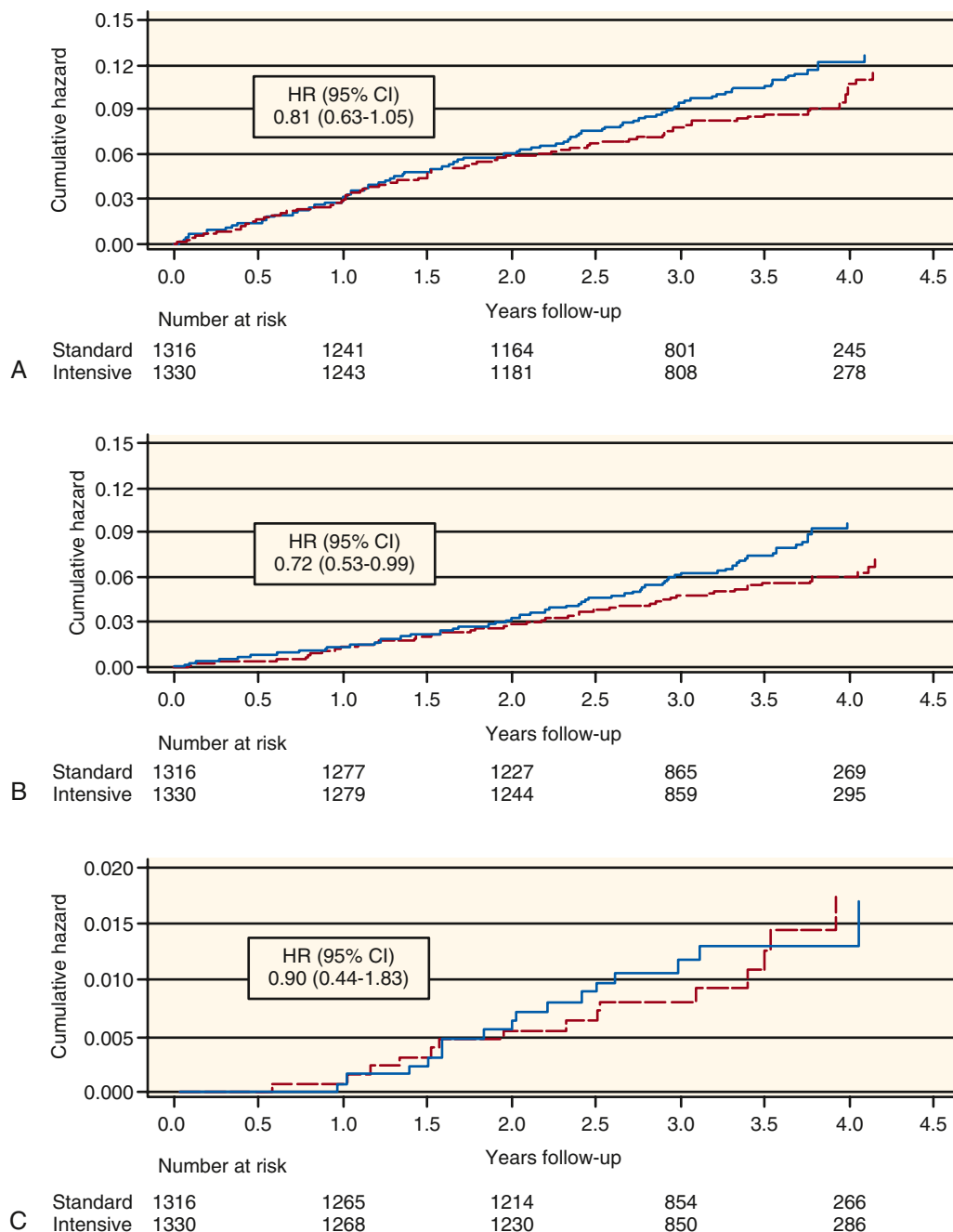


Fig. 54.4 Kaplan-Meier curves for prespecified outcomes in the Systolic Blood Pressure Intervention Trial (SPRINT) participants with chronic kidney disease. (A) The primary cardiovascular outcome, defined as the composite of myocardial infarction, acute coronary syndrome, stroke, acute decompensated heart failure, and death from cardiovascular causes. (B) The all-cause death outcome. (C) The main kidney outcome, defined as the composite of a decrease in estimated glomerular filtration rate of 50% or more from baseline (confirmed by repeat testing ≥ 90 days later) or the development of end-stage kidney disease. The broken lines depict the intensive blood pressure treatment group [systolic blood pressure (SBP) < 120 mm Hg]; the solid lines depict the standard blood pressure treatment group (SBP < 140 mm Hg). CI, Confidence interval; HR, hazard ratio. (From Cheung AK, Rahman M, Reboussin DM, et al. Effects of intensive BP control in CKD. *J Am Soc Nephrol*. 2017;28:2812–2823.)

(IDNT), greater kidney protection was observed in persons who achieved lower blood pressure: Achieved SBP > 149 mm Hg was associated with a 2.2-fold increased risk of developing ESKF or of serum creatinine doubling, in comparison with an achieved SBP < 134 mm Hg, independent of ARB treatment.¹⁰⁵ Progressive lowering of SBP to 120 mm Hg was associated with improved kidney outcomes and improved rates of patient survival, an effect independent of baseline

GFR. However, the lower blood pressure observed was not a primary aim of these studies, and it cannot be assumed that this observed association (between lower blood pressure and improved kidney and patient outcomes) is causative.

Several meta-analyses have combined data from multiple trials to examine the potential benefit of lower BP targets in persons with CKD. In one study that included 11 randomized trials with 1860 participants who had nondiabetic CKD

and proteinuria >1 g/day, the lowest risk of progression was associated with an achieved SBP of 110 to 129 mm Hg, independent of ACE inhibitor treatment.¹⁰⁶ A more recent analysis that included 8127 participants from nine trials (including SPRINT) did not show a significant difference between intensive and usual BP control on the annual rate of change in GFR (MD, 0.07; 95% CI, -0.16 to 0.29 mL/min/year), doubling of serum creatinine concentration or 50% reduction in GFR (RR, 0.99; 95% CI, 0.76–1.29), ESKF (RR, 0.96; 95% CI, 0.78–1.18), composite kidney outcome (RR, 0.99; 95% CI, 0.81–1.21), or all-cause mortality (RR, 0.81; 95% CI, 0.64–1.02). Intensive BP control did reduce mortality (RR, 0.78; 95% CI, 0.61–0.99) in a sensitivity analysis that excluded participants with diabetes. In addition, a trend toward a slower rate of GFR decline was observed in those with proteinuria >1 g/day.¹⁰⁷ By contrast, another recent meta-analysis of 18 randomized trials involving 15,924 participants with and without diabetes reported a 14.0% lower risk of all-cause mortality associated with intensive BP lowering (odds ratio [OR], 0.86; 95% CI, 0.76–0.97; *P* = .01). These findings remained similar when data were reanalyzed without the data from SPRINT.¹⁰⁸ The difference in results between meta-analyses is likely attributable to the inclusion of different trials in each, but they are broadly in keeping with the findings of the individual trials discussed earlier.

By contrast, several sources indicate that excessive lowering of blood pressure may be associated with adverse effects in some persons with CKD. In one meta-analysis,¹⁰⁶ an achieved SBP lower than 110 mm Hg was associated with an increased risk of CKD progression (RR, 2.48; 95% CI, 1.07–5.77), and in IDNT, an achieved SBP lower than 120 mm Hg was associated with increased rates of all-cause mortality and no further improvement in kidney outcomes.¹⁰⁵ In addition, secondary analysis of data from the Ongoing Telmisartan Alone and in Combination with Ramipril Global Endpoint Trial (ONTARGET) revealed that hypertensive persons with risk factors for cardiovascular disease who achieved an SBP of <120 mm Hg had a significantly higher rate of cardiovascular mortality than did those who achieved an SBP of 120 to 129 mm Hg.¹⁰⁹ Similarly, the Action to Control Cardiovascular Risk in Diabetes (ACCORD) blood pressure study reported no difference in primary cardiovascular outcomes (nonfatal myocardial infarction, nonfatal cerebrovascular accident, or cardiovascular death) among persons with diabetes randomly assigned to achieve an SBP target of <120 mm Hg, in comparison with a conventional control target of <140/80 mm Hg. During follow-up, the mean SBP was 119.3 mm Hg in persons undergoing intensive therapy and 133.5 mm Hg in those undergoing standard therapy. Persons undergoing intensive therapy demonstrated a slightly lower stroke rate (annual rates of 0.32% vs. 0.53%) but a higher rate of treatment-related adverse events (3.3% vs. 1.3%). Specifically, there were significantly more instances of an eGFR of <30 mL/min/1.73 m² among persons receiving intensive therapy than among those receiving standard therapy (99 vs. 52 events; *P* < .001), and no difference in the primary cardiovascular composite outcome.¹¹⁰

Current guidelines differ somewhat in their recommendations for blood pressure targets. The KDIGO guidelines recommend a SBP of 120 mm Hg or less for diabetic and nondiabetic adults with stage G1–G4 or A2–A3

CKD who are not on dialysis.¹¹¹ The 2017 American College of Cardiology/American Heart Association guidelines did not adopt the low systolic BP target of <120 mm Hg supported by SPRINT but did recommend a target of <130/80 mm Hg for persons with increased cardiovascular risk, including those with diabetes and/or CKD.¹¹² We support the KDIGO recommendations with a caveat that treatment should be individualized and that low BP targets should be avoided in those at increased risk of adverse effects.

Clinical Relevance

Evidence from the Systolic Blood Pressure Intervention Trial (SPRINT) indicates that a lower target for the treatment of systolic blood pressure may reduce all-cause mortality and reduce cardiovascular risk but may also be associated with some increase in risk of acute kidney injury and hyperkalemia. The risks versus benefits of a lower blood pressure target should be evaluated in each individual and blood pressure targets should be discussed with patients as part of shared decision making.

PHARMACOLOGIC INHIBITION OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Numerous published clinical trials and meta-analyses provide clear evidence to support the use of pharmacologic inhibition of the RAAS with either ACE inhibitors or ARBs as an essential component of any strategy aiming to achieve maximal kidney protection in persons with CKD (Table 54.1). Although ACE inhibitors and ARBs differ significantly in their effects on the RAAS, experimental studies indicate that both treatments produce similar changes in glomerular hemodynamics (for a given blood pressure change) and afford equivalent kidney protection in a variety of CKD models.¹¹³

DIABETIC CHRONIC KIDNEY DISEASE

In 1993, the Captopril Collaborative Study Group published results of the first large prospective RCT to clearly show specific kidney protection attributable to ACE inhibitor treatment, a landmark event in the development of strategies for achieving kidney protection in persons with diabetes and CKD.¹¹⁴ Persons (*n* = 409) with type 1 diabetes and established nephropathy (urine protein excretion >0.5g/day; serum creatinine levels <2.5 mg/dL) were randomly assigned to receive captopril or placebo, and a blood pressure goal of <140/90 mm Hg was set for both groups. After a median follow-up period of 3 years, captopril treatment was associated with a 50% reduction in the risk of the combined endpoint of death, dialysis, and kidney transplantation and a 48% reduction in the risk of serum creatinine doubling. Because blood pressure control was not statistically different between the groups, the additional kidney protection was not attributable simply to the antihypertensive effects of ACE inhibitors.

These results prompted several further studies to investigate whether ACE inhibitors may also benefit persons with early stage nephropathy characterized by microalbuminuria. A meta-analysis of 12 such studies, including 689 persons with type 1 diabetes who were monitored for at least 1 year,

Table 54.1 Summary of Studies Showing the Kidney-Protective Effects of RAAS Inhibitors, SGLT2 Inhibitors, nsMRAs, and GLP1RAs in Diabetic and Nondiabetic Persons With Chronic Kidney Disease

CKD Type	Trial Outcome	Reference
Angiotensin-Converting Enzyme Inhibitors		
Type 1 DM + CKD	↓ Risk of dialysis or death	Lewis et al. ¹¹⁴
Type 1 DM + microalbuminuria	↓ Risk of overt nephropathy	Mathiesen et al. ³⁰¹ Laffel et al. ³⁰² Viberti et al. ³⁰³
Type 1 DM + normoalbuminuria	No significant benefit	Mauer et al. ¹³⁰ EUCLID Study Group ³⁰⁴
Type 2 DM + CKD	Benefit in 1 study only	Bakris et al. ³⁰⁵
Type 2 DM + microalbuminuria	↓ Risk of overt nephropathy	Ravid et al. ^{119,120} Sano et al. ¹¹⁶ Trevisan and Tiengo ¹¹⁶ Agardh et al. ¹¹⁸ Ahmad et al. ¹²¹ Ruggenenti ¹²⁸
Type 2 DM + normoalbuminuria	↓ Risk of developing microalbuminuria	Ravid et al. ¹²²
Nondiabetic CKD	↓ Doubling of creatinine level/ESKF	Ruggenenti et al. ^{138,139}
Angiotensin Receptor Blockers		
Type 1 DM + normoalbuminuria	Small ↑ in albuminuria or no benefit	Mauer et al. ¹³⁰ Bilous et al. ¹³³
Type 2 DM + normoalbuminuria	↓ Risk of developing microalbuminuria (small ↑ in rate of cardiovascular mortality)	Bilous et al. ¹³³ Haller et al. ¹³⁴
Type 2 DM + microalbuminuria	↓ Risk of overt nephropathy	Parving et al. ¹²⁵
Type 2 DM + CKD	↓ Risk of doubling of creatinine level ↓ Risk of ESKF	Lewis et al. ¹²⁴ Brenner et al. ¹²³
Sodium-Glucose Cotransporter 2 Inhibitors		
Type 2 DM and nondiabetic CKD	↓ Risk of CKD progression, ESKF, or death from kidney failure	Perkovic et al. ¹⁷⁰ Heerspink et al. ¹⁷¹ EMPA-KIDNEY ¹⁷³
Nonsteroidal Mineralocorticoid Receptor Antagonist		
Type 2 DM + albuminuria	↓ Risk of CKD progression, death from kidney failure, ESKF or transplantation	Bakris et al. ¹⁶⁰
Glucagon-Like Peptide-1 Receptor Agonist		
Type 2 DM + albuminuria	↓ Risk of CKD progression, death from kidney failure, ESKF or transplantation	Perkovic et al. ¹⁸⁴

CKD, Chronic kidney disease; DM, diabetes mellitus; ESKF, end-stage kidney disease; GLP1RA, glucagon-like peptide-1 receptor agonist;

revealed that ACE inhibitor treatment was associated with a significant reduction in the risk of progression to overt nephropathy (OR, 0.38) and three times the incidence of normalization of microalbuminuria.¹¹⁵

ACE inhibitors also have benefits at earlier stages of nephropathy in people with type 2 diabetes. Several studies, including the diabetic subgroup analysis of the Heart Outcomes Prevention Evaluation (HOPE) study, demonstrated beneficial effects of ACE inhibitor treatment among persons with type 2 diabetes in decreasing microalbuminuria¹¹⁶⁻¹¹⁸ or in reducing the number of persons progressing from microalbuminuria to overt proteinuria (risk reduction, 24% to 67%).¹¹⁹⁻¹²² In addition, the HOPE study reported a 25% reduction in the combined primary endpoint of myocardial infarction, stroke, or

cardiovascular death in ramipril-treated persons with type 2 diabetes and risk factors for cardiovascular disease.

Likewise, three large RCTs published simultaneously established a clear role for ARB therapy in achieving kidney protection for persons with type 2 diabetes. In the Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan (RENAAL) trial, 1513 persons with overt diabetic nephropathy were randomly assigned to receive ARB treatment or placebo and were monitored for a mean of 3.4 years.¹²³ ARB treatment was associated with significant reductions in the incidence of serum creatinine doubling (RR reduction, 25%) and in ESKF (RR reduction, 28%).

In IDNT, 1715 persons with overt diabetic nephropathy were randomly assigned to receive treatment with the ARB irbesartan, amlodipine, or placebo.¹²⁴ After a mean of

2.6 years, the risk of serum creatinine doubling was 33% lower with irbesartan than with placebo and 37% lower with irbesartan than with amlodipine. ARB treatment was associated with a 23% reduction in the risk of ESKF in comparison with placebo and amlodipine, but this reduction was not statistically significant. Of importance is that close matching of achieved blood pressure between groups in both these trials implies that, as with the ACE inhibitor studies, the additional renoprotective effects of ARB treatment could not be attributed merely to their antihypertensive effects.

In a third study, investigators examined the renoprotective effects of an ARB (irbesartan) in 590 persons with type 2 diabetes, hypertension, and microalbuminuria.¹²⁵ Participants were randomly assigned to receive irbesartan at one of two different dosages (300 or 150 mg/day) or placebo. After 2 years, there were significant differences in the incidence of overt proteinuria (5.2%, 9.7%, and 14.9%), and the higher dose of irbesartan was associated with substantial reduction in the risk of overt nephropathy (HR, 0.30; 95% CI, 0.14–0.61) in comparison with placebo. This dose-dependent effect indicates that when ARBs are used to treat diabetes-associated microalbuminuria, the dose should be titrated up to the maximum antihypertensive dose.

A meta-analysis confirmed the results of individual trials by showing significant reductions in the risk of ESKF (RR, 0.78; 95% CI, 0.67–0.91) and in doubling of serum creatinine level (RR, 0.79; 95% CI, 0.67–0.93), as well as a reduction in risk of progression from microalbuminuria to macroalbuminuria (RR, 0.49; 95% CI, 0.32–0.75) among diabetic persons treated with ARB versus placebo.¹²⁶ However, these trials did not demonstrate a reduction in all-cause mortality.

Preventing the onset of diabetic kidney disease

Evidence of kidney protection with RAASi therapy in persons with established diabetic kidney disease prompted further studies to investigate whether early initiation of RAASi treatment could prevent or delay the onset of diabetic kidney disease. Following several trials that reported mixed results,^{122,127–130} three meta-analyses were conducted. One analysis included only studies of type 2 diabetes with albuminuria or proteinuria as an outcome; it revealed statistically significant reductions in albuminuria in association with ACE inhibitor treatment versus placebo.¹³¹ A second, larger analysis combined data from studies of type 1 and type 2 diabetes and revealed a reduced, albeit not statistically significant, risk of serum creatinine doubling (RR, 0.60; 95% CI, 0.34–1.05) or ESKF (RR, 0.64; 95% CI, 0.40–1.03) but stronger evidence of reduced risk of progression of microalbuminuria to macroalbuminuria (RR, 0.45; 95% CI, 0.28–0.71) with ACE inhibitor treatment versus placebo. All-cause mortality was significantly reduced in persons receiving ACE inhibitors (RR, 0.79; 95% CI, 0.63–0.99).¹²⁶ A third meta-analysis involved data from 16 studies of the effect of ACE inhibitor treatment on reducing the risk of microalbuminuria in type 1 and type 2 diabetes. This meta-analysis revealed a significantly reduced risk of developing microalbuminuria with ACE inhibitors in comparison with placebo (RR, 0.60; 95% CI, 0.43–0.84) or calcium channel blocker treatment (RR, 0.58; 95% CI, 0.40–0.84).¹³²

With respect to ARBs, several RCTs have been conducted to investigate the potential role of ARB treatment to prevent the development of microalbuminuria in persons with diabetes. In persons with type 1 diabetes, two trials showed no benefit.^{130,133} In hypertensive persons with type 2 diabetes, treatment with olmesartan was associated with a 23% delay in the time to onset of microalbuminuria versus placebo.¹³⁴ However, a small but significantly higher incidence of death from cardiovascular causes was observed, particularly in persons with a previous history of cardiovascular disease and in those with the greatest reduction in blood pressure.

On the basis of the above data, the KDIGO guidelines recommend treatment with an ACE inhibitor or ARB in all adults with diabetes and urine albumin excretion ≥ 30 mg/day or more (or equivalent).¹³⁵ In people with diabetes, CKD, and normoalbuminuria, the KDIGO guidelines also provide expert opinion to support the use of ACE inhibitors or ARBs for the first-line treatment of hypertension, principally for the established cardioprotective benefits of RAAS inhibition. There is little evidence to suggest a kidney-protective effect of ACE inhibitors or ARBs in people with diabetes, CKD, and normoalbuminuria. For further discussion of the management of diabetic nephropathy, see [Chapter 41](#).

NONDIABETIC CHRONIC KIDNEY DISEASE

After reports of kidney protection with ACE inhibitor treatment in diabetic nephropathy, further studies sought to investigate the renoprotective potential of ACE inhibitors in nondiabetic forms of CKD. One early study demonstrated a 53% reduction in the risk of the composite endpoint (serum creatinine doubling or ESKF) in association with ACE inhibitor treatment, but significantly lower blood pressure in persons receiving ACE inhibitors versus placebo made it impossible to separate the beneficial effects of lowering blood pressure from any unique effects of ACE inhibitor treatment.¹³⁶ By contrast, in the REIN study of 352 persons with nondiabetic CKD and urine protein levels exceeding 1 g/day, similar control of blood pressure was achieved in the participants randomly assigned to receive an ACE inhibitor or placebo. Among persons with at least 3 g/day of urine protein at baseline, the study was stopped early because the rate of decline in GFR in participants receiving the ACE inhibitor was significantly lower (0.53 vs. 0.88 mL/min/month)¹³⁷; further analysis showed a significantly lower risk of the combined endpoint (serum creatinine doubling or ESKF) in the participants taking ACE inhibitors (RR, 1.91; 95% CI, 1.10–3.33 for the placebo recipients).

One-hundred eighty-six persons from the REIN study who had < 3 g/day of urine protein were monitored for a median of 31 months after randomization. In findings similar to those of participants with more severe proteinuria, ACE inhibitor treatment significantly reduced the incidence of ESKF (for placebo recipients, RR, 2.72; 95% CI, 1.22–6.08), particularly among those with a GFR of < 45 mL/min at baseline.¹³⁸ After the randomized phase of the study, persons who had received placebo were switched to ACE inhibitors, and those taking ACE inhibitors continued treatment. In a finding consistent with those of the first phase of the study, there was a significant reduction in the rate of decline in GFR of persons switched to ACE inhibitors. In addition, persons continuing with ACE inhibitor treatment showed a further reduction in the rate of GFR decline. Participants

who had received ACE inhibitors from the start of the REIN study had a significantly lower risk of reaching ESKF than those who switched to ACE inhibitors after the initial phase (for placebo recipients, RR, 1.86; 95% CI, 1.07–3.26). In fact, from 36 to 54 months of follow-up, no additional persons in the former group experienced ESKF.¹³⁹ Of interest is that a small number of persons who continued taking ACE inhibitors exhibited an increase in GFR after prolonged treatment.¹⁴⁰

One RCT confirmed that the kidney-protective benefits of ACE inhibitor treatment may be observed even in advanced stages of CKD. Among 244 persons with a serum creatinine level of 3.1 to 5.0 mg/dL at baseline, treatment with ACE inhibitor versus placebo was associated with a 52% reduction in urine protein levels and a 43% reduction in the risk of the primary endpoint (serum creatinine doubling, ESKF, or death).¹⁴¹ A meta-analysis of 11 studies that included 1860 persons with nondiabetic CKD¹⁴² revealed that ACE inhibitor treatment was associated with significantly lower risks of ESKF (RR, 0.69; 95% CI, 0.51–0.94) and with the composite endpoint of serum creatinine doubling or ESKF (RR, 0.70; 95% CI, 0.55–0.88). Moreover, the benefits of ACE inhibitor treatment were greater in persons with more severe baseline proteinuria but were inconclusive in persons with urine protein levels of <0.5 g/day. A further analysis restricted to persons with autosomal dominant polycystic kidney disease showed greater reduction in proteinuria with ACE inhibitor treatment, but overall evidence of slowing CKD progression was inconclusive and was limited to persons with more severe proteinuria.¹⁴³

In addition to the kidney-protective benefits of ACE inhibitor treatment, the HOPE study reported substantial reductions in overall mortality (RR, 0.84) and cardiovascular mortality (RR, 0.74) among 9297 participants who were at increased risk of cardiovascular disease receiving an ACE inhibitor versus placebo.¹⁴⁴ Although the HOPE study did not include large numbers of persons with nondiabetic CKD, cardiovascular disease remains the most widespread cause of morbidity and mortality among these persons, and the data therefore provide further support for the use of ACE inhibitor therapy in persons with CKD.

With respect to ARBs, no large RCTs have investigated the kidney-protective effects of ARBs versus other antihypertensive drugs in nondiabetic kidney disease. Nonetheless, it is reasonable to expect that ARBs will provide similar kidney-protective benefits in nondiabetic CKD to those observed in diabetic kidney disease.

On the basis of the evidence, the 2024 KDIGO guidelines recommend treatment with an ACE inhibitor or ARBs as the first-line antihypertensive in persons with all forms of CKD and albuminuria >30 mg/g.⁸ In normotensive persons, the KDIGO guidelines similarly recommend treatment with an ACE inhibitor or ARB in adults with CKD and urine albumin excretion exceeding 30 mg/g (or equivalent).⁸

COMBINATION TREATMENT WITH ANGIOTENSIN-CONVERTING ENZYME INHIBITORS AND ANGIOTENSIN RECEPTOR BLOCKERS

The differing mechanisms of ACE inhibitors and ARBs on the RAAS imply that in combination, they may have additive or synergistic effects. However, the added antihypertensive effects of combination therapy have made it difficult to

separate the benefits of additional blood pressure lowering from added kidney protection directly attributable to dual blockade of the RAAS. Only one RCT yielded results indicating benefit associated with ACE inhibitor and ARB combination therapy in CKD with respect to serum creatinine doubling or ESKF incidence. Publication of that study was, however, withdrawn because of concerns about the conduct of the study and integrity of the data.¹⁴⁵

Furthermore, the results of the ONTARGET trial cast doubt on the benefit of dual therapy.¹⁴⁶ In ONTARGET, 25,620 persons with hypertension and additional cardiovascular risk factors were randomly assigned to receive treatment with an ACE inhibitor alone, an ARB alone, or a combination of both.¹⁴⁷ The primary analysis revealed no difference in CVEs between the randomized groups, but the number of events for the composite kidney outcome—need for dialysis, doubling of serum creatinine level, and death¹⁴⁶—was increased with combination treatment (14.5%; HR, 1.09; 95% CI, 1.01–1.18; $P = 0.037$) versus single-agent treatment (ARB: 13.4%; ACE inhibitor 13.5%). This excess was attributable predominantly to more acute dialysis and to the combination of all types of dialysis and serum creatinine doubling. In addition, hyperkalemia ($K > 5.5$ mmol/L) was more frequent with combination treatment (ACE inhibitor, 3.2%; ARB, 3.3%; combination, 5.6%; $P < 0.001$ for combination vs. ACE inhibitor).

A further RCT (VA NEPHRON-D) in 1448 people with type 2 diabetes and UACR of at least 300 mg/g was stopped early due to safety concerns. Analysis of available data found no benefit with respect to the primary outcome of CKD progression, ESKF, or death in participants randomized to combination versus monotherapy. Of concern, those receiving combination treatment showed a significantly higher incidence of AKI (12.2 vs. 6.7 events per 100 person-years; $P < 0.001$) and hyperkalemia (6.3 vs. 2.6 events per 100 person-years; $P < 0.001$).¹⁴⁸

In a meta-analysis of 59 RCTs comparing the efficacy and safety of combination versus single RAASi therapy in CKD, combination treatment was associated with significant improvement in urine albumin and protein excretion, as well as blood pressure control. These beneficial effects, however, were associated with a net 1.8 mL/min/1.73 m² decline in GFR, a significant increase in serum potassium level (3.4% higher rate of hyperkalemia), and a 4.6% higher rate of hypotension. There was no effect on doubling of the serum creatinine level, hospitalization, or mortality.¹⁴⁹

The KDIGO guidelines do not recommend the use of combination ACE inhibitor and ARB therapy for kidney protection¹¹¹ and NICE guidelines,¹⁵⁰ and the 2017 American College of Cardiology/American Heart Association guidelines¹¹² recommend against combination treatment.

MINERALOCORTICOID ANTAGONISM

Aldosterone has been identified as an important factor in the pathogenesis of hypertension, cardiovascular disease, and progressive kidney injury through hemodynamic and profibrotic actions. Treatment with steroidal mineralocorticoid receptor antagonists (sMRAs) has produced kidney-protective effects in experimental¹⁵¹ and small clinical studies.¹⁵² One meta-analysis of their use in comparison with other RAAS inhibitors included data

from 10 trials and 845 participants. In comparison with ACE inhibitor or ARB plus placebo, a nonselective MRA (spironolactone) added to ACE inhibitor or ARB treatment significantly reduced proteinuria (weighted MD, -0.80 g/day; 95% CI, -1.23 to -0.38) and blood pressure, but these developments did not translate into an improvement in GFR (weighted MD, -0.70 mL/min/ 1.73 m²; 95% CI, -4.73 to 3.34). There was a significant increase in the risk of hyperkalemia with the addition of an sMRA (RR, 3.06; 95% CI, 1.26–7.41).¹⁵³ A more recent meta-analysis of 27 studies confirmed that treatment with spironolactone alone or in combination with an ACE inhibitor or ARB was associated with reduction in proteinuria (standardized MD, -0.61 g/day; 95% CI, -1.08 to -0.13) and blood pressure (MD, -3.44 mm Hg; 95% CI, -5.05 to -1.83) but provided no data on reduction in CVEs or ESKF. In addition, spironolactone was associated with an increased risk of hyperkalemia (RR, 2.00; 95% CI, 1.25–3.20) and gynecomastia compared with ACE inhibitor or ARB (or both) (RR, 5.14; 95% CI, 1.14–23.23).¹⁵⁴ The kidney-protective potential of eplerenone, a more selective sMRA, has also been evaluated. In one RCT that enrolled 336 persons with hypertension, UACR 30 to 599 mg/g, and eGFR of 50 mL/min/ 1.73 m² or above, treatment with eplerenone for 52 weeks, added to ACE inhibitor or ARB therapy, resulted in a 17.3% decrease in UACR, whereas a 10.3% increase was observed in those who received placebo ($P = .02$). Serum potassium was higher with eplerenone treatment, but no episodes of severe hyperkalemia (serum potassium > 5.5 mmol/L) were observed.¹⁵⁵

Given the off-target steroidal side effects of sMRAs, nonsteroidal (ns)MRAs were developed as selective and potent alternatives to block aldosterone. Currently, the only widely available nsMRA with proven kidney and cardiovascular benefits is finerenone, although other nsMRAs such as esaxerenone are approved for use only in Japan. In comparison with sMRAs such as spironolactone and eplerenone, finerenone exhibits greater selectivity for the mineralocorticoid receptor and is at least as potent as spironolactone in reducing binding of aldosterone to the mineralocorticoid receptor.^{156–158} Due to the bulky chemical structure of finerenone, it also exerts more potent blocking of mineralocorticoid receptor cofactor binding as compared with spironolactone and eplerenone; which may reduce downstream gene expression of proinflammatory and profibrotic genes. In addition, with respect to side effects, finerenone has minimal affinity for androgen receptors and does not produce untoward side effects such as gynecomastia. Additionally, dedicated head-to-head RCTs have shown that finerenone is associated with a lower incidence of hyperkalemia. The phase II mineralocorticoid Receptor Antagonist Tolerability Study (ARTS) compared the effect of finerenone with placebo and spironolactone in people with heart failure with reduced ejection fraction and moderate CKD and showed that the incidence of hyperkalemia (serum potassium ≥ 5.6 meq/L) was lower with finerenone as compared with spironolactone (3.7% vs. 12.7% respectively, $P = 0.03$). Furthermore, the incidence of investigator-reported kidney failure was lower with finerenone versus spironolactone (1.5% vs. 7.9%, $P = 0.04$).¹⁵⁹

With respect to clinical outcomes, finerenone reduces the risk of CKD progression. The Effect of Finerenone

on Chronic Kidney Disease Outcomes in Type 2 Diabetes (FIDELIO-DKD) trial compared the effect of finerenone versus placebo on the primary composite of kidney failure, a sustained decrease of $\geq 40\%$ in the eGFR from baseline, or death from renal causes.¹⁶⁰ The study included people with type 2 diabetes and CKD, and the CKD eligibility criteria included people with a UACR of 30 to 300 mg/g, an eGFR of 25 to 60 mL/min/ 1.73 m², and diabetic retinopathy, or people with a UACR 300 to 5000 mg/g and an eGFR of 25 to 75 mL per minute per 1.73 m². Finerenone reduced the primary kidney composite endpoint by 18% versus placebo (HR 0.82, 95% CI 0.73–0.93), albeit with a 1.4% absolute increase in the risk of hyperkalemia-related discontinuation of study medication (2.3% for finerenone vs. 0.9% for placebo). Similar evidence of the kidney-protective benefit of finerenone was noted in the companion cardiovascular pivotal RCT, FIGARO-DKD, which compared finerenone to placebo in people with type 2 diabetes and CKD.¹⁶¹ In FIGARO-DKD, the CKD eligibility criteria included people with a UACR of 30 to 300 mg/g and an eGFR of 25 to 90 mL/min/ 1.73 m² or people with a UACR of 300 to 5000 mg/g and an eGFR ≥ 60 mL/min/ 1.73 m². Finerenone reduced the prespecified secondary kidney composite outcome of kidney failure, a sustained decrease of $\geq 40\%$ in the eGFR from baseline, or death from renal causes by 13% (HR 0.87, 95% CI: 0.76–1.01) in people with type 2 diabetes and CKD. A pooled analysis of FIDELIO-DKD and FIGARO-DKD reinforced the finding for the kidney composite of kidney failure, a sustained decrease of $\geq 40\%$ in the eGFR from baseline, or death from renal causes, as well as for hard outcomes such as ESKF (HR 0.80, 95% CI: 0.64–0.99; Fig. 54.5).¹⁶² Only a small minority of participants in these trials were also treated with an SGLT2 inhibitor, so it remains uncertain whether the benefits of treatment with finerenone are additive to those of SGLT2 inhibitors (discussed later).

Given consistent evidence of efficacy in FIDELIO-DKD and FIGARO-DKD, the 2022 KDIGO guidelines for diabetes management in CKD recommend an nsMRA with proven kidney or cardiovascular benefit for people with type 2 diabetes, an eGFR ≥ 25 mL/min/ 1.73 m², normal serum potassium, and UACR ≥ 30 mg/g despite a maximal tolerated dose of RAS inhibitor.

An alternative strategy for aldosterone antagonism involves limiting its production in the zona glomerulosa of the adrenal gland through inhibition of aldosterone synthase, which is the enzyme that catalyzes the final rate-limiting steps for producing aldosterone from deoxycorticosterone. However, a challenge in the development of aldosterone synthase inhibitors (ASIs) is the similarity in the amino acid sequence between aldosterone synthase and cortisol synthase and the potential for ASIs to inhibit cortisol production and increase the risk of adrenal insufficiency. Two highly selective ASIs—baxdrostat and lorundrostat—have been studied in phase II RCTs focused on their blood pressure lowering effect and both agents have demonstrated efficacy in blood pressure lowering. Furthermore, a phase II RCT involved 586 people with CKD was completed for the highly selective ASI BI 690517, alone or in combination with empagliflozin, a sodium-glucose cotransporter-2 (SGLT2) inhibitor. BI 690517 administered at the 10-mg dose reduced UACR by 39% versus 3% with placebo.¹⁶³ There was no increased risk of adrenal insufficiency. The phase III

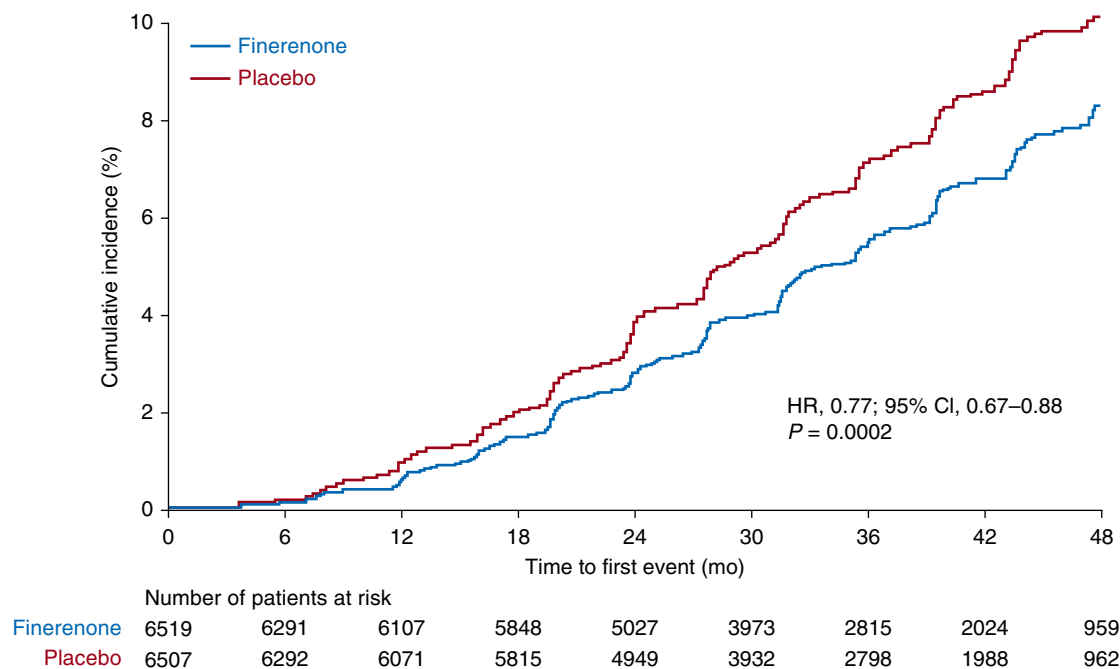


Fig. 54.5 Results of a pooled analysis of the FIDELIO-DKD and FIGARO-DKD clinical trials of finerenone versus placebo in people with diabetic kidney disease. The analysis included 13,026 persons with a median follow-up of 3 years and showed that treatment with finerenone was associated with a 23% reduction in the cumulative incidence of the composite kidney outcome of time to kidney failure, sustained 57% or more decrease in eGFR from baseline, or kidney death. (From Bakris GL, Ruilope LM, Anker SD, et al. A prespecified exploratory analysis from FIDELITY examined finerenone use and kidney outcomes in patients with chronic kidney disease and type 2 diabetes. *Kidney Int.* 2023;103:196–206.)

RCT of BI 690517, entitled EASi-KIDNEY, is under way and additional phase II RCTs of baxdrostat (NCT05432167) and larundrostat are ongoing to examine the efficacy of ASIs with respect to managing CKD.

DIRECT RENIN INHIBITORS

Direct renin inhibitors block the RAAS at its rate-limiting step (the conversion of angiotensinogen to angiotensin I) and may therefore achieve more complete blockade of the RAAS than do ACE inhibitors or ARBs. Direct renin inhibitors (e.g., aliskiren) are effective antihypertensive drugs and reduced proteinuria in experimental models of CKD^{164,165} and RCTs.^{166,167} However, a large randomized trial that included 8561 people with type 2 diabetes and albuminuria or cardiovascular disease was stopped prematurely after the second interim efficacy analysis. Despite greater lowering of blood pressure and albuminuria with a combination of direct renin inhibitor and ARB therapy, no benefit was observed with respect to the composite primary endpoint of CVE, CKD progression, ESKF, or death versus ARB monotherapy, but combination therapy was associated with a higher incidence of hyperkalemia and hypotension.^{168,169} There is currently no evidence that direct renin inhibitor therapy, alone or in combination with ACE inhibitor therapy, affords more effective kidney protection than ACE inhibitor or ARB therapy alone.

SODIUM-GLUCOSE COTRANSPORTER 2 (SGLT2) INHIBITORS

SGLT2 inhibitors block the reabsorption of glucose and sodium in the proximal tubule, thereby increasing delivery of these solutes to the macula densa, stimulating

tubuloglomerular feedback and afferent arteriolar vasoconstriction. The net effect of these changes is a decrease in intraglomerular pressure, which manifests as an acute dip of 3 to 5 mL/min/1.73 m² in eGFR following the initiation of SGLT2 inhibitors. Within 6 weeks of initiating SGLT2 inhibitors, eGFR stabilizes and the rate of eGFR decline slows, thereby affording kidney protection compared with placebo-treated patients. Other effects of SGLT2 inhibitors include improved glycemic control, reduced blood pressure, amelioration of arterial stiffness, and weight loss, although the kidney-protective effects of SGLT2 inhibitors are independent of metabolic benefits.

In dedicated kidney RCTs, SGLT2 inhibitors reduce the risk of hard clinical endpoints including CKD progression, ESKF, and reduction of albuminuria down to an eGFR of 20 mL/min/1.73 m². To date, there have been three pivotal kidney RCTs of SGLT2 inhibitors, all of which were stopped early for efficacy. The CREDENCE trial enrolled 4401 adults with type 2 diabetes, eGFR 30 to <90 mL/min/1.73 m², and UACR 300 to 5000 mg/g despite treatment with the maximum dose of an ACE inhibitor or ARB. The risk of the composite primary outcome of ESKF, doubling of serum creatinine, and cardiovascular or renal death was reduced by 30% in participants who received canagliflozin (HR, 0.70; 95% CI, 0.59–0.82; $P < 0.001$), and the risk of ESKF was reduced by 32% (HR, 0.68; 95% CI, 0.54–0.86; $P = 0.002$).¹⁷⁰ The geometric mean UACR was also reduced by 31% (95% CI, 26%–35%). Likewise, the DAPA-CKD trial enrolled 4304 adults with and without type 2 diabetes, with an eGFR of 25 to 75 mL/min/1.73 m², UACR of 200 to 5000 mg/g on a stable dose of an ACE inhibitor or ARB.¹⁷¹ The risk of the primary composite of a sustained decline in eGFR of $\geq 50\%$, ESKF, or death from renal or cardiovascular causes was

reduced by 39% in participants who received dapagliflozin versus placebo (HR 0.61, 95% CI, 0.51–0.72, $P < 0.001$), and the risk of ESKF was reduced by 36% (HR, 0.64; 95% CI, 0.50–0.82). The geometric mean UACR was reduced by 29.3% (95% CI, 25.2%–33.1%).¹⁷² The EMPA-KIDNEY trial enrolled 6609 adults with and without type 2 diabetes, with an eGFR of 20 to <45 mL/min/1.73 m² or an eGFR of 45 to 90 mL/min/1.73 m² with a UACR of ≥ 200 mg/g.¹⁷³ The risk of the composite primary outcome of progression of kidney disease (defined as ESKF, a sustained decrease in eGFR to <10 mL/min/1.73 m², a sustained decrease in eGFR of $\geq 40\%$ from baseline, or death from renal causes) or death from cardiovascular causes was reduced by 28% (HR, 0.72, 95% CI, 0.64–0.82), and the risk of ESKF was reduced by 33% (HR 0.67; 95% CI 0.52–0.85). In meta-analyses of the pivotal RCTs, SGLT2 inhibitor treatment reduced the risk of kidney disease progression by 37% (relative risk (RR) 0.63, 95% CI 0.58–0.69) and also reduced the risk of acute kidney injury by 23% (RR 0.77, 95% CI 0.70–0.84), cardiovascular death or hospitalization for heart failure by 23% (RR 0.77, 95% CI 0.74–0.81) and cardiovascular death by 14% (RR 0.86, 95% CI 0.81–0.92). The treatment effects were consistent in people with and without diabetes and irrespective of the underlying cause of kidney disease (Fig. 54.6).¹⁷⁴ The treatment effects were also consistent across

baseline eGFR or albuminuria.¹⁷⁵ On an absolute basis, in adults with CKD, prevention of 1 kidney disease progression event can be achieved by treating only 91 adults with type 2 diabetes for 1 year and 67 adults without diabetes for 1 year with an SGLT2 inhibitor versus placebo.¹⁷⁴ When combined with RAAS inhibitors, treatment with an SGLT2 inhibitor versus placebo also results in an additional 7.9 years without doubling serum creatinine or kidney failure in people with albuminuria without diabetes.¹⁷⁶ Finally, at a population level, screening of adults 35 to 75 years old for albuminuria followed by treatment with an SGLT2 inhibitor would prevent ESKF requiring dialysis in 398,000 to 658,000 people over their lifetime, depending on the frequency of screening, and is cost-effective.¹⁷⁷ To date, there are no dedicated kidney outcome studies of persons without diabetes who have the earliest stages of CKD (eGFR ≥ 45 mL/min/1.73 m² with UACR < 200 mg/g). There are also limited data in people with single kidneys, polycystic kidney disease, or kidney transplants in whom SGLT2 inhibitor treatment is currently not indicated. Studies are forthcoming in people with type 1 diabetes who have concurrent CKD (NCT06217302).

SGLT2 inhibitors have a favorable side effect profile. Common side effects include low blood pressure and volume depletion. Before starting these medications, it is relevant to

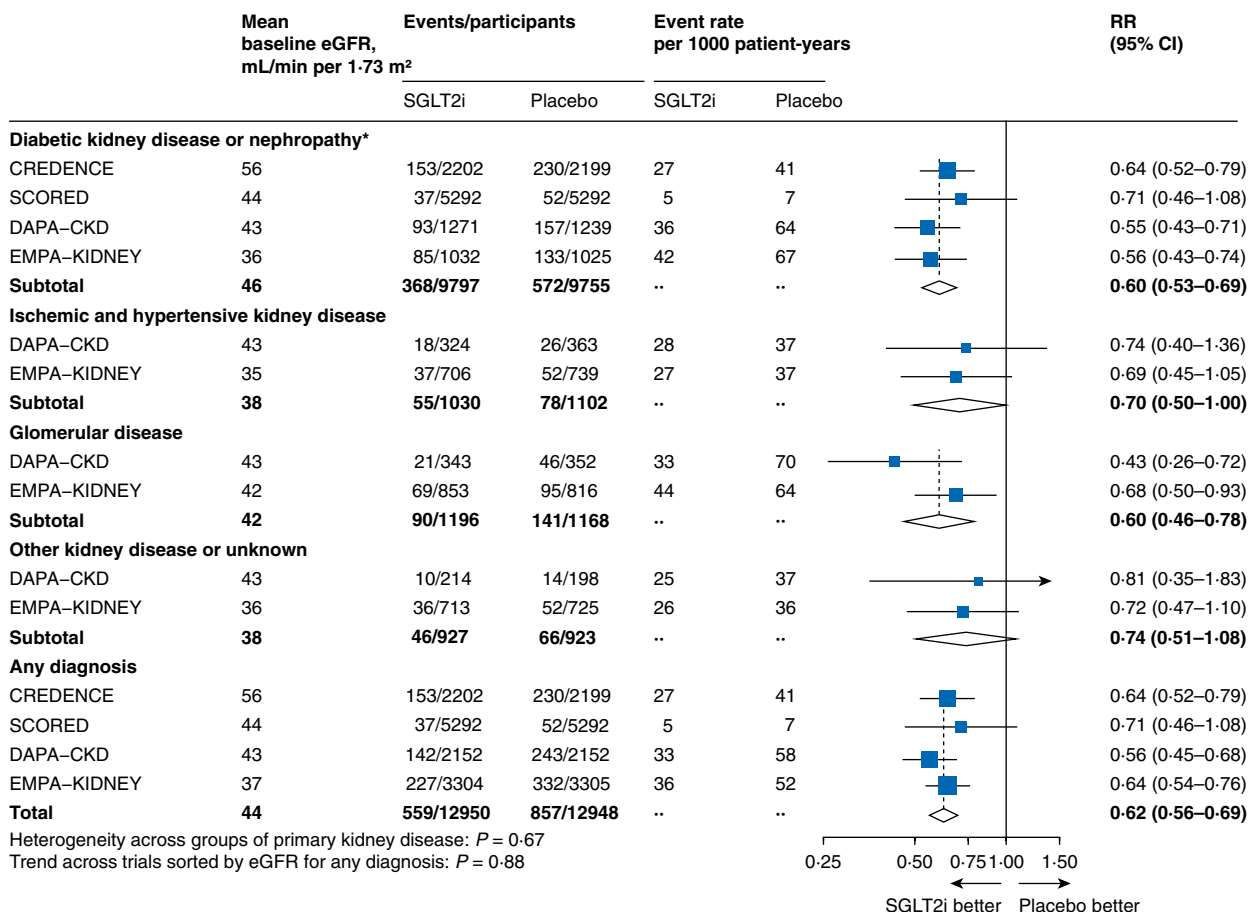


Fig. 54.6 Results of a meta-analysis of 13 randomized trials and 90,413 people of SGLT2 inhibitors versus placebo on the outcome of kidney disease progression. The analysis shows that the effect of SGLT2 inhibitors versus placebo is consistent, irrespective of the cause of primary kidney disease. (From Nuffield Department of Population Health Renal Studies Group; SGLT2 Inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium. Impact of diabetes on the effects of sodium glucose co-transporter-2 inhibitors on kidney outcomes: collaborative meta-analysis of large placebo-controlled trials. *Lancet*. 2022;400(10365):1788–1801.)

make sure patients are not hypotensive or hypovolemia and to decrease or stop concomitant medications that have no benefit on hard clinical endpoints beyond blood pressure lowering, such as diuretics or calcium channel blockers. The risk of genital mycotic infection is increased with SGLT2 inhibitors, although this can be readily managed through counseling regarding the need for routine genital hygiene and the use of topical antifungals if needed. In the event of an infection, prescription of antifungal treatment is usually sufficient, without discontinuation of the SGLT2 inhibitor. However, if the infection is refractory to treatment, a temporary discontinuation of the SGLT2 inhibitor can be considered, followed by reinitiation at the same or a lower dose. There is no meaningful increase in the risk of urinary tract infections with SGLT2 inhibitors.^{174,178} Less common but important side effects include diabetic ketoacidosis (HR 2.12; 95% CI, 1.49–3.04), although the absolute risk increase in people with type 2 diabetes is marginal, ranging from 0.3 to 1.5 events per 1000 person-years.¹⁷⁴ In contrast, the risk of DKA is higher in people with type 1 diabetes, ranging from 4 to 7%, and this has been a barrier to the use of SGLT2 inhibitors in this population.¹⁷⁹ Potential risk factors for DKA associated with SGLT2 inhibitor use include large changes in insulin dose, dietary carbohydrate restriction, surgical stress or intercurrent illness impairing oral intake, lean body habitus, and latent autoimmune diabetes in adults.¹⁸⁰ Clinicians can therefore mitigate the risk of DKA through gradual dose titration of insulin dose, avoiding $\geq 20\%$ one-time insulin dose reduction, discontinuation of SGLT2 inhibitors during acute illness or before elective surgery, and counseling of patients regarding the risk of DKA.

Given the favorable benefit-risk profile of SGLT2 inhibitors, the KDIGO diabetes guidelines recommend initiating SGLT2 inhibitors for preventing CKD progression and albuminuria lowering in persons with type 2 diabetes and an eGFR ≥ 20 mL/min/1.73 m², irrespective of UACR. For people without diabetes, the 2024 KDIGO guidelines for the management of CKD recommend initiation of an SGLT2 inhibitor in persons with an eGFR ≥ 20 mL/min/1.73 m² and a UACR ≥ 200 mg/g or in people with CKD, an eGFR 20 to 45 mL/min/1.73 m², or heart failure, irrespective of UACR.⁸ In persons stabilized on SGLT2 inhibition, these medications can be continued even after eGFR falls below 20 mL/min/1.73 m² and used until ESKF, KRT, or eGFR < 10 mL/min/1.73 m².

GLUCAGON-LIKE PEPTIDE-1 RECEPTOR AGONISTS AND GLUCOSE-DEPENDENT INSULINOTROPIC POLYPEPTIDE RECEPTOR AGONISTS

GLP1RAs are long-acting analogs of GLP1, a naturally occurring incretin hormone that is principally secreted from the enteroendocrine L cells of the distal ileum and colon in response to food intake.¹⁸¹ GLP1 exerts insulinotropic effects through receptors on pancreatic β cells, and in the context of diabetes, postprandial production of GLP1 is attenuated.¹⁸¹ GLP1RAs mimic the effect of GLP1 and agonize the GLP1 receptor in the pancreas and therefore stimulate endogenous insulin production with the net benefit of improving glycemic control. GLP1RA agents were initially developed as treatments for type 2 diabetes,

although due to the presence of GLP1 receptors in multiple organ beds including the heart, vasculature, kidney, and central nervous system, these medications exert multiple systemic effects including weight loss as discussed earlier. Experimental studies have identified that GLP1RA treatment has kidney-protective effects in diabetic and nondiabetic animal models, at least in part due to anti-inflammatory effects mediated by inhibition of receptor for advanced glycation endproducts (RAGE).¹⁸²

With respect to CKD, the effect of GLP1RAs on the kidney was first demonstrated in dedicated kidney-specific analysis of pivotal cardiovascular outcome trials in persons with type 2 diabetes. In a meta-analysis of 6 cardiovascular outcome RCTs, involving 44,378 participants, GLP1RAs reduced the incidence of the composite kidney endpoint of development of macroalbuminuria, doubling of serum creatinine or at least 40% decline in eGFR, KRT, or death due to kidney disease by 21% as compared with placebo (HR 0.79; 95% CI, 0.73–0.87).¹⁸³ GLP1RAs also reduced the incidence of the composite of doubling of serum creatinine or $\geq 40\%$ decline in eGFR, by 14% compared with placebo (HR 0.86; 95% CI 0.72–1.02).¹⁸³ These hypothesis-generating findings prompted the initiation of an RCT with primary kidney endpoints, the “Evaluate Renal Function with Semaglutide Once Weekly” (FLOW) trial, which compared semaglutide to placebo in people with type 2 diabetes and CKD, defined as an eGFR between 50 and 75 mL/min/1.73 m² with a UACR between 300 and 5000 mg/g or an eGFR between 25 and 50 mL/min/1.73 m² and UACR between 100 and 5000 mg/g.¹⁸⁴ The primary outcome of FLOW was a kidney composite of an eGFR decline $\geq 50\%$, ESKF, or death from kidney disease or cardiovascular disease.¹⁸⁴ The FLOW trial was stopped early for efficacy after a median follow-up of 3.4 years and reported a 24% reduction in the risk of the primary endpoint (HR 0.76; 95% CI, 0.66–0.88), as well as a 21% reduction in a secondary, kidney-specific endpoint (primary endpoint without cardiovascular death), 18% reduction in risk of first major CVE, and 20% reduction in all-cause mortality (Fig. 54.7).¹⁸⁴ The results were consistent across all subgroups including SGLT2i use at baseline, though the latter included only 15.6% of participants.¹⁸⁴ GLP1RAs may also have kidney-protective effects in persons without diabetes. The “Semaglutide Effects on Cardiovascular Outcomes in People with Overweight or Obesity” (SELECT) trial enrolled persons who were overweight or obese but without diabetes (21.3% with CKD).¹⁸⁵ In this population, treatment with semaglutide treatment was associated with a 20% reduction in the primary endpoint of major CVEs.¹⁸⁵ A prespecified secondary analysis reported a 22% reduction in the composite kidney endpoint of progression to eGFR < 15 mL/min/1.73 m², initiation of KRT, death due to kidney disease, eGFR decline of $\geq 50\%$, or progression to uACR > 300 mg/g (HR 0.78; 95% CI, 0.63–0.96).¹⁸⁶ Further trials with primary kidney endpoints are therefore warranted to specifically evaluate the kidney-protective effects of GLP1RAs in persons without diabetes.

Alongside GLP-1, GIP is currently the only other naturally incretin hormone. GIP is produced in the intestinal K-cells, which are principally located in the duodenum and proximal jejunum. Similar to GLP1, GIP is released in response to food intake and in healthy individuals, GIP

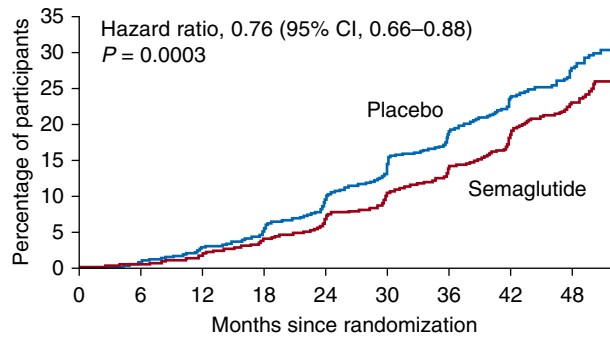


Fig. 54.7 Cumulative incidence graph demonstrating the effect of semaglutide versus placebo in 3533 people in the FLOW trial on the primary outcome of an eGFR decline $\geq 50\%$, kidney failure, or death from kidney disease or cardiovascular disease. (From Perkovic V, Tuttle KR, Rossing P, et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med*. 2024;391(2):109–121.)

exerts its insulinotropic effect through receptors located on pancreatic β cells.¹⁸¹ In people with type 2 diabetes, mechanistic studies suggest there is no acute insulinotropic response to infusion of GIP.¹⁸⁷ However, in a dedicated well-powered RCT, tirzepatide, a coagonist of the GLP-1 and GIP receptor, was superior to the selective GLP1RA semaglutide in reducing HbA1c in people with type 2 diabetes.¹⁸⁸ There is also early evidence that tirzepatide may confer kidney protection. The SURPASS-4 RCT compared the effect of tirzepatide to insulin glargine in people with type 2 diabetes, BMI ≥ 25 kg/m², and established cardiovascular disease or high risk of CVEs. In a post hoc analysis focused on kidney outcomes, tirzepatide reduced the incidence of the composite kidney endpoint of eGFR decline $\geq 40\%$, ESKF, death from kidney failure, or new-onset macroalbuminuria by 42% as compared with insulin glargine (HR 0.58, 95% CI 0.43–0.80).¹⁸⁹ Dedicated kidney-specific RCTs of tirzepatide are needed to definitively establish its efficacy for kidney protection.

Clinical Relevance

Treatment with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers are “first line” in persons with albuminuria to achieve kidney protection. Sodium-glucose cotransporter 2 inhibitor (SGLT2i) treatment should be added for all persons with type 2 diabetes, CKD, and eGFR >20 mL/min/1.73 m². For persons with CKD without diabetes, SGLT2 inhibitors should be added for those with eGFR 20 to 45 mL/min/1.73 m², irrespective of albuminuria or ≥ 45 mL/min/1.73 m² with albuminuria. For persons with type 2 diabetes and residual albuminuria, a nsMRA (finerenone) and a glucagon-like peptide-1 receptor agonist (GLP1RA)-like semaglutide should be added, unless contraindicated. Treatment with these medications should be started sequentially at a low dose and titrated upward. Serum creatinine and potassium should be checked approximately 2 weeks after initiation or dose titration. Additional monitoring is generally not required for GLP1-RA initiation.

MANAGEMENT OF HYPERURICEMIA

A number of studies have highlighted evidence that elevated serum uric acid concentration is a risk factor for the development and progression of CKD.^{190,191} It has also been associated with excess cardiovascular risk and hypertension in a cohort of community-based persons.¹⁹²

An early small randomized trial of allopurinol in 113 persons with CKD reported a 47% reduction in risk of CKD progression (defined as decrease in eGFR of >0.2 mL/min/month) in participants treated with allopurinol, as well as a 71% reduction in risk of new CVEs and 62% reduction in risk of hospitalization.¹⁹³ Long-term follow-up of these participants demonstrated further benefit with a 68% reduction in risk of a kidney event (initiation of RRT or 50% reduction in GFR or doubling of serum creatinine) and a 57% reduction in risk of CVE.¹⁹⁴ However, two subsequent large RCTs have found no evidence of benefit of serum urate lowering. First, the PERL trial randomized 530 people with type 1 diabetes, an eGFR of 40 to 99.9 mL/min/1.73 m², and a serum urate of 267 μ mol/L or higher to allopurinol versus placebo. Although allopurinol reduced serum urate versus the placebo group, the mean between-group difference in iohexol-based measured GFR was only -0.6 mL/min/1.73 m².¹⁹⁵ Likewise, the CKD-FIX randomized people with stage 3 or 4 CKD and no history of gout to allopurinol versus placebo. Eligible participants had an eGFR UACR of ≥ 265 mg/g or an eGFR decrease of at least 3 mL/min/1.73 m² in the year preceding randomization. The study randomized 369 people, short of the intended target of 620 people, because of slow recruitment. There was no effect of allopurinol versus placebo in the change in eGFR from randomization to week 104.¹⁹⁶ Furthermore, a meta-analysis that included 29 placebo-controlled trials of allopurinol, febuxostat, and other uric acid-lowering treatments (4471 participants with CKD) reported that all were effective in lowering serum uric acid concentration but only allopurinol improved GFR and none of the treatments lowered proteinuria.¹⁹⁷ Therefore we do not recommend uric acid lowering in persons without gout as part of the management of CKD.

TREATMENT OF METABOLIC ACIDOSIS

As the number of functioning nephrons declines, CKD leads to net retention of hydrogen ions, which begins when GFR falls below 40 to 50 mL/min/1.73 m².¹⁹⁸ Among persons in whom GFR decreases from 90 to <20 mL/min/1.73 m², the prevalence of metabolic acidosis rises from 2% to 39% and is higher among younger persons and those with diabetes.¹⁹⁹ As the patient approaches ESKF, the plasma bicarbonate concentration tends to stabilize between 15 and 20 mEq/L. Chronic metabolic acidosis has multiple adverse consequences, including increased protein catabolism, increased bone turnover, induction of inflammatory mediators, insulin resistance, and increased production of corticosteroids and parathyroid hormone. Several observational studies, including the CRIC study, have identified low serum bicarbonate as a risk factor for CKD progression,²⁰⁰ although post hoc analysis of data

from persons with diabetic kidney disease enrolled in the RENAAL and IDNT studies found that an association between lower serum bicarbonate and kidney outcomes was not maintained after adjustment for baseline GFR.²⁰¹

The first study to show convincing kidney protection with bicarbonate supplementation was from a single center and involved 134 persons with advanced CKD (creatinine clearance rates between 15 and 30 mL/min/1.73 m²) and baseline serum bicarbonate concentrations of 16 to 20 mEq/L. The participants were randomly assigned to receive treatment with oral bicarbonate or no treatment.²⁰² After 2 years of follow-up, there was a lower mean rate of decline in creatinine clearance (1.88 vs. 5.93 mL/min/1.73 m²) and a lower risk of ESKF among the persons who received the bicarbonate treatment than among the controls (6.5% vs. 33%). In a subsequent randomized, placebo-controlled trial in persons with a mean eGFR of 75 mL/min/1.73 m², treatment with sodium bicarbonate for 5 years was associated with a slower rate of decline in eGFR (derived from plasma cystatin C measurements) compared with placebo or treatment with sodium chloride.²⁰³ Western diets are typically acid producing, but the addition of significant portions of fruits and vegetables can move this to a base-producing state. Further studies have reported that correction of acidosis with a diet rich in fruits and vegetables was as effective as sodium bicarbonate in ameliorating kidney damage in early (stage 1 or 2 CKD)²⁰⁴ and more advanced (stage 4 CKD) disease.²⁰⁵ Furthermore, a recent trial has reported benefits even in people with mild acidosis. People with stage 3 CKD and serum bicarbonate 22 to 24 mmol/L were randomized to oral bicarbonate supplementation, a diet rich in fruits and vegetables, or “usual care.” All participants received treatment with an RAAS inhibitor, and SBP was controlled to lower than 130 mm Hg. Both interventions achieved an increase in serum bicarbonate and were associated with a decrease in urinary angiotensinogen. After 3 years, both interventions were associated with less albuminuria and GFR decline than the “usual care” group.²⁰⁶ The KDIGO guidelines recommend bicarbonate supplementation using pharmacologic treatment, with or without dietary interventions, to prevent serum bicarbonate levels below 18 mEq/L,⁸ but further studies are required to further investigate whether this may also be beneficial in the setting of less severe acidosis.

MONITORING AND SAFETY CONSIDERATIONS

Regular monitoring is essential for optimizing therapeutic interventions to slow CKD progression and ensure safety. Kidney function is best assessed with the use of eGFR, derived from a serum creatinine measurement with cystatin C where available, using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation, recommended by KDIGO.⁸ This strategy allows direct monitoring of the rate of GFR decline and assessment of the therapeutic goal to reduce this decline to <1 mL/min/year, a rate associated with normal aging. Most laboratories now facilitate this monitoring by reporting eGFR with every serum creatinine measurement. In addition, monitoring allows for the detection of side effects of drug treatment, particularly electrolyte disorders (hyperkalemia and hyponatremia), as well as acute changes in kidney function related to volume

depletion. The appropriate frequency for monitoring of persons with CKD should depend on the CKD stage, previous rate of GFR decline, risk of future GFR decline, and use of medication that may cause acute deteriorations in GFR or electrolyte disorders (especially RAAS inhibitors, diuretics, SGLT2is, and MRAs).

INITIATION AND TITRATION OF RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM INHIBITORS

Despite clear trial evidence of the kidney-protective and cardioprotective effects of ACE inhibitors, ARBs, and nonsteroidal MRAs, some physicians remain hesitant about prescribing these medications to persons with stages 3 and 4 CKD. This caution results from concerns about kidney dysfunction induced by these drugs, with a potential rise in serum creatinine or potassium level (reviewed by Schoolwerth and colleagues²⁰⁷ and Palmer²⁰⁸).

The initiation of therapy with RAAS inhibition may provoke an acute decline in GFR or hyperkalemia in persons with CKD, particularly in those with volume depletion, poor cardiac status, elderly persons, those with stage 4 or 5 disease, and persons with atherosclerotic renovascular disease. GFR and electrolytes should therefore be checked before and approximately 1 to 2 weeks after treatment is started or dosage is increased. On the basis of a review (but not a meta-analysis) of data for 12 RCTs, it has previously been recommended that an acute increase in serum creatinine of up to 30% after initiation of RAAS inhibitor therapy is acceptable because it is attributable to glomerular hemodynamic effects and correlates with slower subsequent annual decline in GFR.²⁰⁹ This advice has been questioned by analysis of data from a large database of 122,363 persons initiating RAAS inhibitor therapy in primary care. The analysis confirmed that a >30% increase in serum creatinine within 2 months after initiation of RAAS inhibitors (observed in 1.7% of participants) was associated with a significant increase in risks of ESKF, myocardial infarction, cardiac failure, and death, although the incidence rate ratio decreased with time after starting treatment. Importantly, the study also reported a graduated relationship between all of these adverse outcomes and acute increase in serum creatinine at all values ≥10% (Fig. 54.8).²¹⁰ An important limitation of this analysis is that data on proteinuria were not available. Nevertheless, these data indicate that a more cautious approach is warranted. A rapid initial rise in serum creatinine level or a more gradual progressive increase should prompt discontinuation of therapy and consideration of further investigation to exclude renovascular disease (see Chapter 47). It should be remembered that AKI can occur even if RAAS inhibition therapy has been successful for months or years, usually when provoked by factors such as volume depletion or nephrotoxic medications.

ACE inhibitor or ARB treatment should be started at a low dose and titrated upward, with monitoring of creatinine and potassium levels 1 to 2 weeks after each increase. To avoid compromise from intravascular volume depletion, persons should be counseled to omit ACE inhibitor or ARB treatment during vomiting or diarrheal illnesses and to seek medical advice if these illnesses do not resolve within 48 hours. Likewise, it is important to ensure adequate hydration, omit or reduce diuretics for 48 to 72 hours if clinically appropriate, and avoid nonsteroidal anti-inflammatory drugs (NSAIDs)

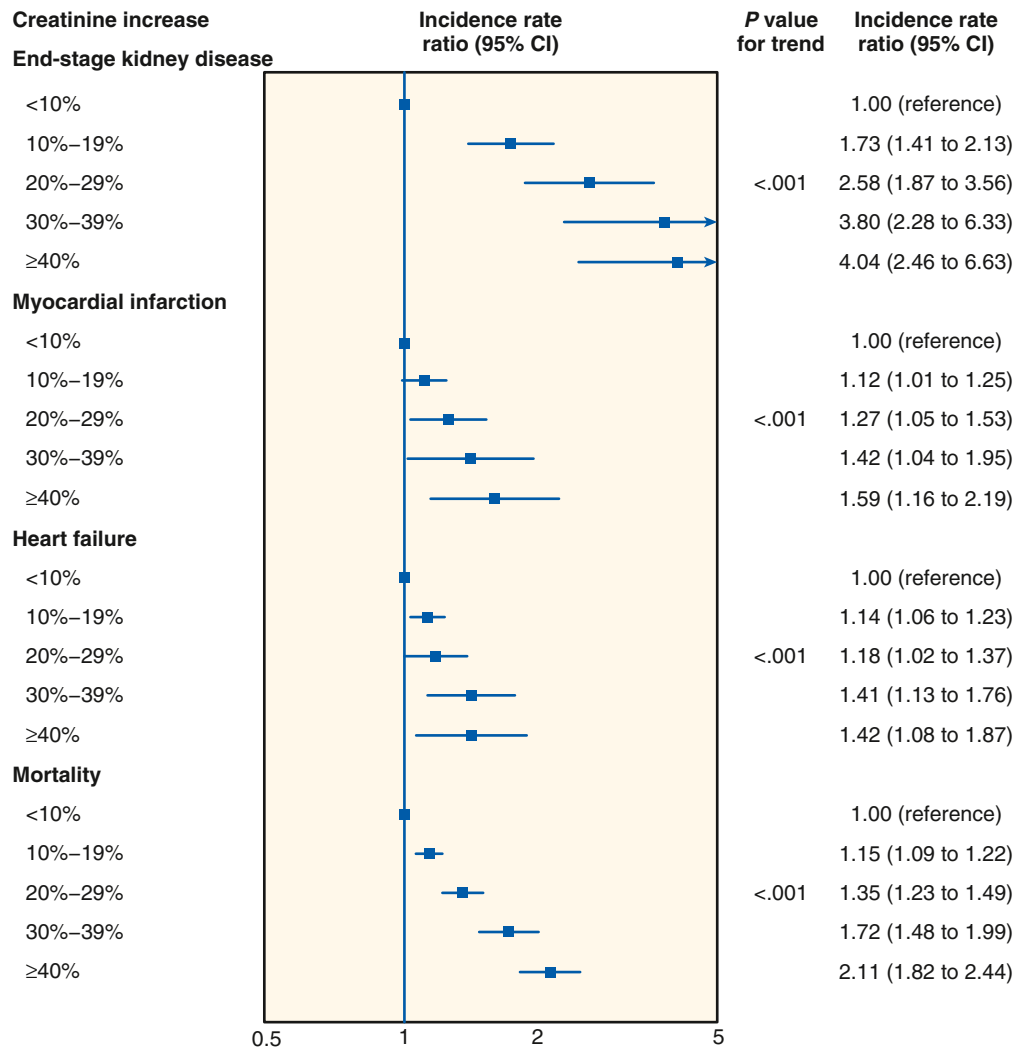


Fig. 54.8 Cardiorenal risks associated with levels of serum creatinine increase within 2 months after initiating treatment with renin-angiotensin-aldosterone system inhibitors. *CI*, Confidence interval. (From Schmidt M, Mansfield KE, Bhaskaran K, et al. Serum creatinine elevation after renin-angiotensin system blockade and long term cardiorenal risks: cohort study. *BMJ*. 2017;356:791.)

before starting an RAAS inhibitor. In general, we strongly advise discontinuation of NSAIDs because these are potent causes of AKI in persons with CKD. General guidance on risk factors for AKI and frequency of monitoring are given in [Tables 54.2 and 54.3](#).

INITIATION AND TITRATION OF SGLT2 INHIBITORS

Initiation of SGLT2 inhibitors is associated with an initial dip in eGFR, but this change reflects a decrease in intraglomerular pressure due to afferent arteriolar vasoconstriction. This initial dip in eGFR generally should not prompt discontinuation of an SGLT2 inhibitor. First, the efficacy of SGLT2 inhibitors on hard kidney outcomes is consistent, irrespective of the magnitude of the initial dip in eGFR.^{211–213} Second, SGLT2 inhibitors reduce the incidence of AKI and hyperkalemia, suggesting that this initial dip does not lead to long-term deleterious effects. In the SGLT2 inhibitor Meta-Analysis Cardio-Renal Trialists' Consortium (SMART-C) meta-analysis, which includes 13 RCTs and 90,313 participants, SGLT2 inhibitors reduced the incidence of AKI by 23% (HR 0.77, 95% CI, 0.70–0.84).

Likewise, in a meta-analysis of 6 RCTs, involving 49875 participants, SGLT2 inhibitors reduced the incidence of serum potassium ≥ 6 mmol/L by 16% (HR 0.84; 95% CI, 0.76–0.93; [Fig. 54.9](#)).¹⁷⁴ This benefit was consistent irrespective of baseline kidney function, history of heart failure, and concomitant medications including RAAS inhibitors or diuretics.²¹⁴ SGLT2 inhibitors may therefore be considered as an adjunctive treatment to facilitate ongoing treatment with RAAS inhibitors when hyperkalemia is a concern. With respect to laboratory monitoring of kidney function, there is no indication for routine monitoring of kidney function or electrolytes in most persons starting an SGLT2 inhibitor except in persons at high risk of AKI such as the elderly, people with BP $< 120/70$ mm Hg, signs/symptoms of volume depletion, or people taking high-dose diuretics.²¹¹

MANAGEMENT OF HYPERKALEMIA INDUCED BY INHIBITORS OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Serum potassium should be assessed and trends across all previous test results considered before initiating RAAS

Table 54.2 Overview of Chronic Kidney Disease Management by Stage

Features	Stages 1 and 2	Stage 3a	Stage 3b	Stage 4	Stage 5
Estimated GFR	≥60 mL/min/1.73 m ² + albuminuria or hematuria or structural kidney damage	45-59 mL/ min/1.73 m ²	30-44 mL/ min/1.73 m ²	15-29 mL/ min/1.73 m ²	<15 mL/min/1.73 m ²
Frequency of monitoring	Annual	6-12 months	3-6 months	3-4 months	1-3 months
Laboratory testing	Annual electrolytes and estimated GFR Annual urine ACR (or other estimate of proteinuria) Baseline anemia and mineral and bone profiles Glucose, lipids, and HbA _{1c} See Table 54.3 for causes of AKI after initiation of ACE inhibitor or ARB therapy.				Check electrolytes and estimated GFR 1 week after new use or higher doses of ACE inhibitors or ARBs; otherwise, assess electrolytes/estimated GFR, mineral and bone, and anemia profiles every 3-6 months, depending on GFR decline.
Blood pressure targets	BP target <130/80 mm Hg with proteinuria BP target <140/90 mm Hg without proteinuria if no clinical or radiologic evidence of ARVD or previous episodes of AKI				Risk of AKI is increased in elderly persons (>75 years), those with CHF, and those with ARVD; 140/90 mm Hg may be more appropriate for these groups.
Blood pressure agents	ACE inhibitor or ARB if urine ACR ≥30 mg/g Most persons need 2-4 agents in total to achieve these targets, in a combination of ACE inhibitors or ARBs and 1 or more of the following: a diuretic (all classes), a calcium channel blocker, and a β-blocker.				Loop diuretics are now usually required for BP and edema control.
SGLT2 Inhibitors	SGLT2 inhibitors in all people with CKD and type 2 diabetes and eGFR ≥20 mL/ min/1.73 m ² SGLT2 inhibitors in all people with CKD without diabetes and an eGFR between 20 and 45 mL/min/1.73 m ² irrespective of UACR or an eGFR of ≥45 mL/min/1.73 m ² with a UACR ≥200 mg/g				Continue SGLT2 inhibitor.
nsMRA	nsMRA for all people with CKD with type 2 diabetes and an eGFR ≥25 mL/ min/1.73 m ² , normal serum potassium concentration, and a UACR ≥30 mg/g despite maximum tolerated dose of ACE inhibitor or ARB				Continue nsMRA as tolerated.
Cardiovascular prevention	Statin if CVD risk ≥10% over 10 years HbA _{1c} <7.0% unless at risk for severe hypoglycemia GLP1RA as the preferred second-line glucose lowering medication for people with diabetes at high cardiovascular risk			Consider statin for all persons. GLP1RA for people with diabetes at high cardiovascular risk	Continue statin.
Bone and anemia complications	If PTH level rises progressively, commence phosphate restriction and then consider therapy with vitamin D or analog. If anemia is out of keeping with GFR, confirm or rule out gastrointestinal blood loss.				Give intravenous iron before ESA if hemoglobin count <10 g/dL. Maintain target hemoglobin count of 10-11.5 g/dL.
Lifestyle and nutritional management	Smoking cessation Moderate exercise up to 30-60 min/day 4-7 days/wk Target weight with BMI <25 kg/m ² Reduced salt intake as per DASH diet <5 g/day GLP1RA as the preferred second line glucose lowering medication for people with diabetes and requiring weight management				Limit dietary potassium excess. Weigh at each clinic and assess fluid overload, anorexia, physical function.
Specific RRT planning steps	Education regarding progression and role of conservative management regarding blood pressure targets and specific primary kidney disease treatment if indicated Hepatitis B vaccination if risk of progression is high			Education on RRT types and palliative care if CKD is progressing or patient is at high risk of progression. Hepatitis B vaccination.	AVF creation PD catheter insertion Enter on list for transplant.

Continued on following page

Table 54.2 Overview of Chronic Kidney Disease Management by Stage (Continued)

Features	Stages 1 and 2	Stage 3a	Stage 3b	Stage 4	Stage 5
Referral guidance from primary care physician to nephrologist	Progressive or abrupt fall in estimated GFR Proteinuria (urine protein levels >0.5 g/day; ACR >300 mg/g or >30 mg/mmol)			Refer unless patient is terminally ill	Refer unless patient is terminally ill.

ACE, Angiotensin-converting enzyme; ACR, albumin-to-creatinine ratio; AKI, acute kidney injury; ARB, angiotensin receptor blocker; ARVD, atherosclerotic renovascular disease; AVF, arteriovenous fistula; BMI, body mass index; BP, blood pressure; CHF, congestive heart failure; CKD, chronic kidney disease; CVD, cardiovascular disease; DASH, Dietary Approaches to Stop Hypertension; ESA, erythropoietin-stimulating agent; GFR, glomerular filtration rate; GLP1RA, glucagon-like peptide 1 receptor agonists; HbA_{1c}, hemoglobin A_{1c}; PD, peritoneal dialysis; nsMRA, nonsteroidal Mineralocorticoid Receptor Antagonist; PTH, parathyroid hormone; RAAS, renin-angiotensin-aldosterone system; RRT, renal replacement therapy; SGLT2, sodium-glucose cotransporter-2.

Table 54.3 Causes of Acute Kidney Injury After Initiation of Therapy With Angiotensin-Converting Enzyme Inhibitor or Angiotensin Receptor Blocker**Blood Pressure Insufficient for Adequate Renal Perfusion**

Poor cardiac output
Low systemic vascular resistance (e.g., as in sepsis)
Volume depletion (gastrointestinal loss, poor oral intake, excess diuretic use)

Presence of Renal Vascular Disease^a

Bilateral renal artery stenosis
Stenosis of dominant or single kidney
Afferent arteriolar narrowing (caused by hypertension and cyclosporine)
Diffuse atherosclerosis in smaller renal vessels

Vasoconstrictor Agents (Nonsteroidal Anti-inflammatory Drugs, Cyclosporine)

^aClinical features of renal vascular disease include vascular bruits (areas of the epigastric, femoral, and carotid arteries), prior rise in serum creatinine level of more than 30%, fall in estimated glomerular filtration rate of more than 20% after beginning of treatment with an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker, and a history of flash pulmonary edema.

inhibitor treatment. If hyperkalemia has been present, the first step to control this is to reduce dietary potassium intake, if necessary, with the assistance of a dietician. If dietary intervention is unsuccessful, novel potassium binders may be considered as an adjunct to prevent hyperkalemia and facilitate ongoing RAAS inhibitor treatment in people with recurrent or persistent hyperkalemia on RAAS inhibitors despite discontinuation of potassium supplements, adherence to a low potassium diet, and concomitant use of an SGLT2 inhibitor (Fig. 54.10).⁶ Historically, the only available potassium binders were sodium polystyrene sulfonate (Kayexalate) and calcium polystyrene sulfonate.

These medications were associated with an increased risk of gastrointestinal injury and gut necrosis,²¹⁵⁻²¹⁷ which resulted in a black box warning by the U.S. Food and Drug Administration. Two novel potassium binders—sodium zirconium cyclosilicate (ZS-9) and patiromer—have since received regulatory approval based on RCTs that established their efficacy and safety. ZS-9 is an insoluble and nonabsorbable compound with a crystalline lattice structure that traps potassium in exchange for sodium and hydrogen. ZS-9 is more selective for potassium over calcium by >25 fold, and its potassium binding capacity is 9-fold greater than sodium polystyrene. In three phase III RCTs of persons with serum potassium >5 mmol/L, of whom 60% to 80% had CKD at baseline, the 10-g dose of ZS-9 significantly reduced serum potassium by 0.7 to 1.3 mmol/L compared with placebo. Normokalemia was maintained during maintenance treatment and hypokalemia was rare.²¹⁸⁻²²⁰ From a tolerability perspective, ZS-9 is associated with an increased risk of edema, likely because potassium is bound in exchange for sodium.²²¹ The clinical implications, if any, of this increased sodium load have yet to be studied.

Patiromer is an insoluble nonabsorbable compound with a spherical bead structure that traps potassium in the colon exchange for calcium. Importantly, patiromer has minimal effect on serum calcium or other markers of mineral metabolism.²²² In phase II and III RCTs, patiromer has demonstrated efficacy in lowering serum potassium and maintaining normokalemia.^{223,224} The phase III OPAL-HK trial assigned 237 participants with stage 3 or 4 CKD, an eGFR of 15 to 60 mL/min/1.73 m², and a serum potassium of 5.1 to 6.5 mmol/L to a 4-week single-group treatment phase with patiromer at an initial dose of 4.2 or 8.4 g twice a day, followed by an 8-week randomized placebo-controlled withdrawal phase in a subset of participants in who had an initial response during the treatment phase.²²⁵ After the initial single-group treatment phase, 76% of participants achieved a serum potassium of 3.8 to <5.1 mmol/L. In the randomized withdrawal phase, recurrence of hyperkalemia, defined as a serum potassium of ≥5.5 mmol/L, occurred in 60% of the participants in the placebo group as compared with 15% in the patiromer group. Constipation was the most common side effect with the use of patiromer, occurring

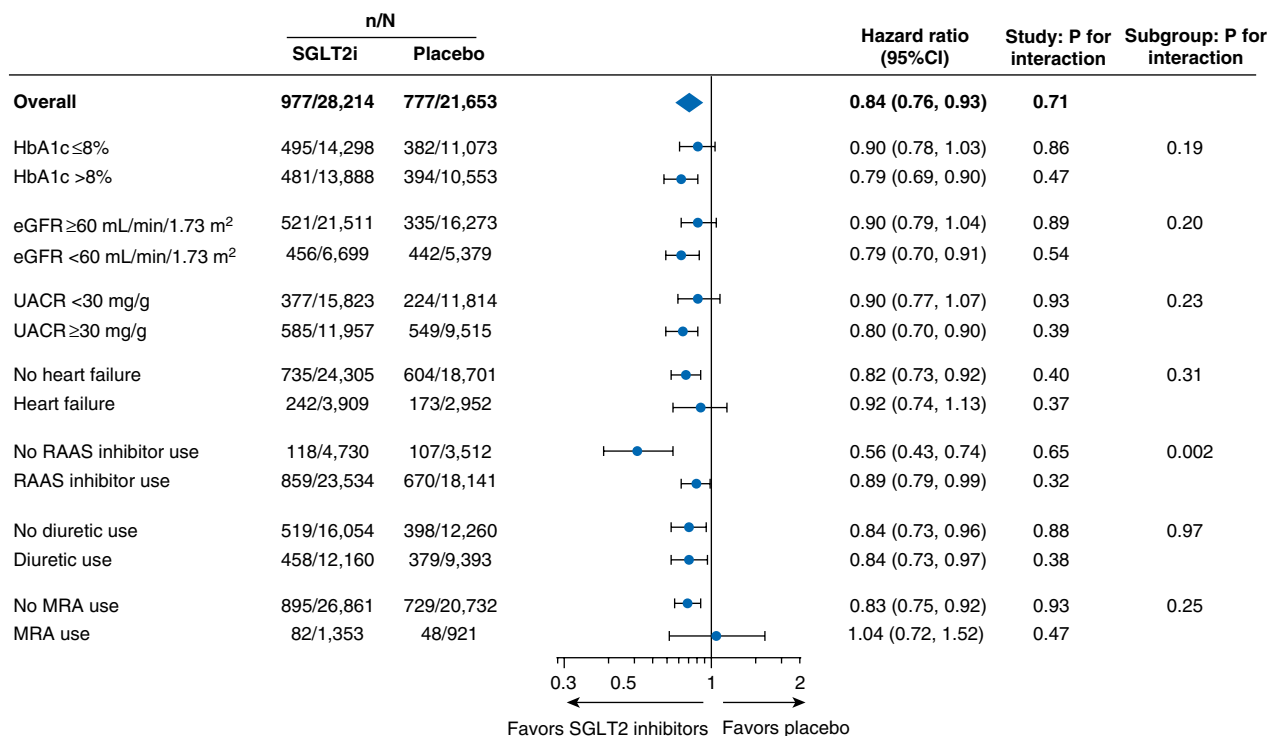


Fig. 54.9 Forest plot showing the results of a meta-analysis of 6 randomized trials involving 49,875 people of SGLT2 inhibitors versus placebo on the outcome of serious hyperkalemia, defined as a central laboratory determined serum potassium ≥ 6 mmol/L. The effect of SGLT2 inhibitors versus placebo on serious hyperkalemia is consistent, irrespective of the baseline characteristics of participants. *eGFR*, Estimated glomerular filtration rate; *HbA1c*, glycohemoglobin; *MRA*, mineralocorticoid receptor antagonist; *RAAS*, renin-angiotensin-aldosterone system; *SGLT2i*, sodium-glucose cotransporter 2 inhibitor; *UACR*, urinary albumin-to-creatinine ratio. (From Neuen BL, Oshima M, Agarwal R, et al. Sodium-glucose cotransporter 2 inhibitors and risk of hyperkalemia in people with type 2 diabetes: a meta-analysis of individual participant data from randomized, controlled trials. *Circulation*. 2022;145(19):1460–1470.)

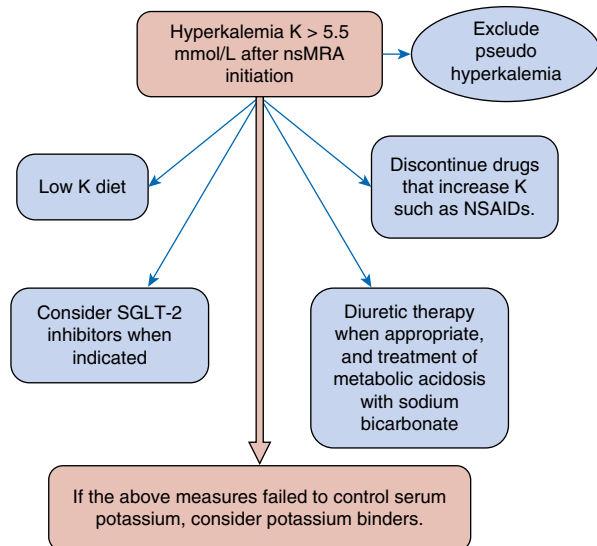


Fig. 54.10 A treatment algorithm for management of nsMRA related hyperkalemia. *NSAID*, Nonsteroidal anti-inflammatory drug; *nsMRA*, nonsteroidal mineralocorticoid receptor antagonist; *SGLT-2*, sodium-glucose cotransporter 2. (From Albakr RB, Sridhar VS, Cheney DZ. Novel therapies in diabetic kidney disease and risk of hyperkalemia: a review of the evidence from clinical trials. *Am J Kidney Dis*. 2023;82(6):737–742.)

in 11% of participants. Hypokalemia occurred in 3% of participants. ZS-9 and patiomer should therefore be considered important adjuncts to manage hyperkalemia and support ongoing use of RAAS inhibitors in people with CKD.

If nonpharmacologic and pharmacologic strategies have been exhausted, discontinuation of RAAS inhibitors can be considered. However, real-world analyses have highlighted the importance of stopping these medications as a last resort since kidney and cardiovascular risks increase when these medications are stopped.²²⁶

While there are no exact estimates of the incidence of hyperkalemia requiring medication discontinuation in the era of combined use of ACE inhibitors, ARBs, SGLT2 inhibitors, and nonsteroidal MRAs, it is reasonable to anticipate that the incidence of hyperkalemia in usual clinical practice will be higher than in RCTs. Participants in RCTs were highly selected and have a lower risk of hyperkalemia than is observed in the general population. This important issue was highlighted by the observation of higher rates of hyperkalemia after publication of results of the Randomized Aldactone Evaluation Study (RALES), in which spironolactone was used in addition to ACE inhibitors for heart failure.^{227,228} The risks associated with hyperkalemia and hypokalemia have been investigated in a large meta-analysis that combined data from 27 general population and CKD cohort studies to include 1,217,986

participants. The lowest risk of adverse outcomes was observed when serum potassium was in the range of 4.0 to 4.5 mmol/L. Compared with a reference value of 4.2 mmol/L, serum potassium concentration >5.5 mmol/L or <3.0 mmol/L was associated with significantly increased risk of all-cause mortality (HR, 1.22; 95% 1.15–1.29 and HR, 1.49; 95% CI, 1.26–1.76, respectively). A similar pattern was observed for the outcomes of cardiovascular mortality and ESKF.²²⁹

STRATEGIES FOR MAXIMAL KIDNEY PROTECTION

PROTEINURIA AS A THERAPEUTIC TARGET

Proteinuria is a marker of glomerular filtration barrier integrity and the magnitude of proteinuria has therefore been used as an indicator of the severity of glomerulopathy. This view has been confirmed by several observations that the severity of proteinuria at baseline is the most important independent predictor of kidney outcomes in randomized trials of persons with diabetic nephropathy^{230,231} and nondiabetic nephropathy.^{232–234} In addition, it has been proposed that proteinuria per se contributes to progressive kidney injury²³⁵ and that amelioration of proteinuria should be viewed as a therapeutic goal.

Support for this hypothesis is derived from many sources. In the MDRD study, reduction in urine protein levels, independent of blood pressure, was associated with slower progression of CKD, and the degree of benefit achieved through blood pressure lowering was dependent on the extent of baseline proteinuria.⁹⁶ Furthermore, several other investigators have observed that the percentage reduction in urine protein level after initiation of ACE inhibitor or ARB treatment and the magnitude of proteinuria during treatment (residual proteinuria) are strong independent predictors of the subsequent rate of decline in GFR among persons with diabetic and nondiabetic CKD.^{230,231} A meta-analysis that included data from 1860 persons with nondiabetic CKD confirmed these findings and showed that during antihypertensive treatment, the achieved level of urine protein was a powerful predictor of the combined endpoint of serum creatinine doubling or onset of ESKF (RR, 5.6 for each 1.0-g/day increase in achieved level of proteinuria).¹⁴² A further meta-analysis of 21 randomized trials of drug treatment in CKD that included 78,342 participants found that for each 30% initial reduction in albuminuria on treatment, the risk of ESKF decreased by 23.7% (95% CI, 11.4%–34.2%) independent of the class of drug used for treatment.²³⁶

One RCT provided direct evidence that the extent to which proteinuria is lowered determines subsequent prognosis. In the Renoprotection of Optimal Antiproteinuric Doses (ROAD) study, 360 nondiabetic persons with proteinuria and CKD were randomly assigned to receive a conventional dosage of benazepril (10 mg/day), a conventional dosage of losartan (50 mg/day), an upward titration of benazepril (range of 10–40 mg/day), or an upward titration of losartan (range of 50–200 mg/day).²³⁷ In upward titration, the dosage of the RAAS-inhibiting agent was increased to maximize the antiproteinuric effect. After a median of 3.7

years, titration of benazepril and losartan to the maximum antiproteinuric dose, in comparison with the fixed conventional dosages, reduced the risk of serum creatinine doubling by 49% and 50%, respectively, and the risk of ESKF by 47% in both groups (Fig. 54.11). Both agents provided similar overall RR reductions at optimal antiproteinuric dosages. Reduction of urine protein levels at 3 months (approximately 50% decrease with the upward-titration strategy) was closely correlated with the subsequent rate of GFR decline. Participants in all groups of the study had similar reductions in blood pressure. A meta-analysis of data from more than 27 million participants has emphasized the continuous increase in risk of all adverse outcomes with any increase in albuminuria, even within the range regarded as “normal.”²³⁸ Regardless of whether proteinuria contributes directly to kidney injury, the strong association between achieved reduction in urine protein and kidney protection in clinical studies implies that amelioration of proteinuria should be regarded as an important therapeutic goal in kidney-protective strategies. We recommend escalation of kidney-protective treatment to reduce proteinuria to the lowest possible magnitude, aiming for a uACR of <30 mg/g (<3 mg/mmol) if possible.²³⁹

Clinical Relevance

Proteinuria should be regarded as a therapeutic target with the aim of reducing albuminuria to <30 mg/day. All kidney-protective treatments including RAAS inhibitor therapy, SGLT2 inhibitors, nsMRAs, and GLP-1RAs should be initiated and escalated to guideline-recommended doses. Antihypertensive therapy should be optimized to achieve guideline-recommended blood pressure targets.

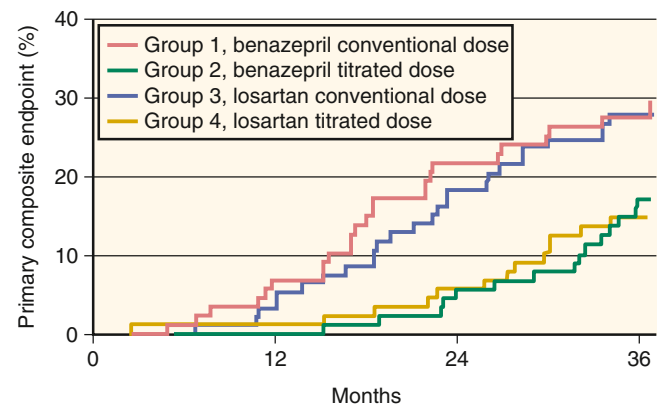


Fig. 54.11 Kaplan-Meier curves showing the incidence of the combined endpoint of doubling of serum creatinine level or end-stage kidney disease and urine protein levels exceeding 1 g/day. Persons in group 1 were randomly assigned to receive standard doses of angiotensin-converting enzyme (ACE) inhibitors; those in group 3 received standard doses of angiotensin receptor blockers (ARBs); those in group 2 received ACE inhibitors titrated upward to the maximum antiproteinuric levels; and those in group 4 received ARBs titrated upward similarly. (From Gruppo Italiano di Studi Epidemiologici in Nefrologia [GISEN]. Randomised placebo-controlled trial of effect of ramipril on decline in glomerular filtration rate and risk of terminal renal failure in proteinuric, non-diabetic nephropathy. *Lancet*. 1997;349:1857–1863.)

TIME COURSE FOR PROTEINURIA RESPONSE

In most persons, the reduction of proteinuria takes several weeks to achieve its maximal effect, which should be considered when dosages are titrated.²⁴⁰ In persons in earlier stages of CKD (with diabetes and microalbuminuria, for instance) the effects are observed more promptly and tend to parallel the time course and fall in blood pressure.²⁴⁰

GOAL-DIRECTED MEDICAL TREATMENT TO ACHIEVE REMISSION OF CHRONIC KIDNEY DISEASE

With the advent of multiple therapeutic options for preventing CKD progression, the adoption of a goal-directed medical treatment strategy that combines the use of ACE inhibitors or ARBs, SGLT2 inhibitors, GLP1RA, and nsMRAs will be important for disrupting the multiple pathophysiologic changes that contribute to CKD (Figs. 54.2 and 54.12). The 2024 KDIGO guidelines for the management of CKD identify ACE inhibitors, ARBs, and SGLT2 inhibitors as the key pillars of treatment to be considered in all people with CKD.⁸ ACE inhibitors or ARBs should be initiated in all people with CKD and UACR ≥ 30 mg/g. Likewise, SGLT2 inhibitors should be considered in all people with CKD and type 2 diabetes with an eGFR ≥ 20 mL/min/1.73 m². In people without diabetes, SGLT2 inhibitors should be initiated in people with eGFR between 20 and 45 mL/min/1.73 m², irrespective of UACR, and in people with an eGFR >45 mL/min/1.73 m² with a UACR ≥ 200 mg/g. ACE inhibitors, ARBs, and SGLT2 inhibitors should also be combined for their kidney-protective benefits.

The addition of nsMRAs should be informed by the presence of residual albuminuria and considered for

additional kidney and cardiovascular protection in all people with type 2 diabetes, an eGFR ≥ 25 mL/min/1.73 m², normal serum potassium concentration, and a UACR ≥ 30 mg/g despite maximum tolerated dose of ACE inhibitor or ARB. GLP1RAs are currently identified in the 2024 KDIGO guidelines as the preferred second-line glucose-lowering agents for the management of glycemia, after metformin and SGLT2 inhibitors. GLP1RAs should be particularly prioritized in people with type 2 diabetes, CKD, and either obesity or established cardiovascular disease, given their efficacy for weight management and reducing ischemic CVEs. The results of the FLOW trial indicate that GLP1RA treatment should now also be considered as an additional pillar in kidney-protective strategies for people with type 2 diabetes and CKD with albuminuria. In people without diabetes, there is currently limited evidence on the kidney-protective effects of nsMRAs and GLP1RA for kidney protection, though further trials are under way for nsMRAs.²⁴¹

The combined effect of SGLT2 inhibitors, GLP1RAs, and nsMRAs on CKD progression has yet to be established. However, each class of medications works through distinct pathways, suggesting that their beneficial effects are likely to be synergistic as opposed to duplicative. In a small crossover RCT involving 46 people with an eGFR of 30 to 90 mL/min/1.73 m² and UACR ≥ 100 mg/g, participants were randomized to a 4-week treatment period with dapagliflozin 10 mg/day, eplerenone 50 mg/day or their combination in random order. The mean percentage reduction in UACR was 19% with dapagliflozin, 34% with eplerenone, and 53% with dapagliflozin combined with eplerenone.²⁴² Meta-analyses of SGLT2 inhibitors and GLP1RAs have also shown that the combined use of these medication classes results in additive

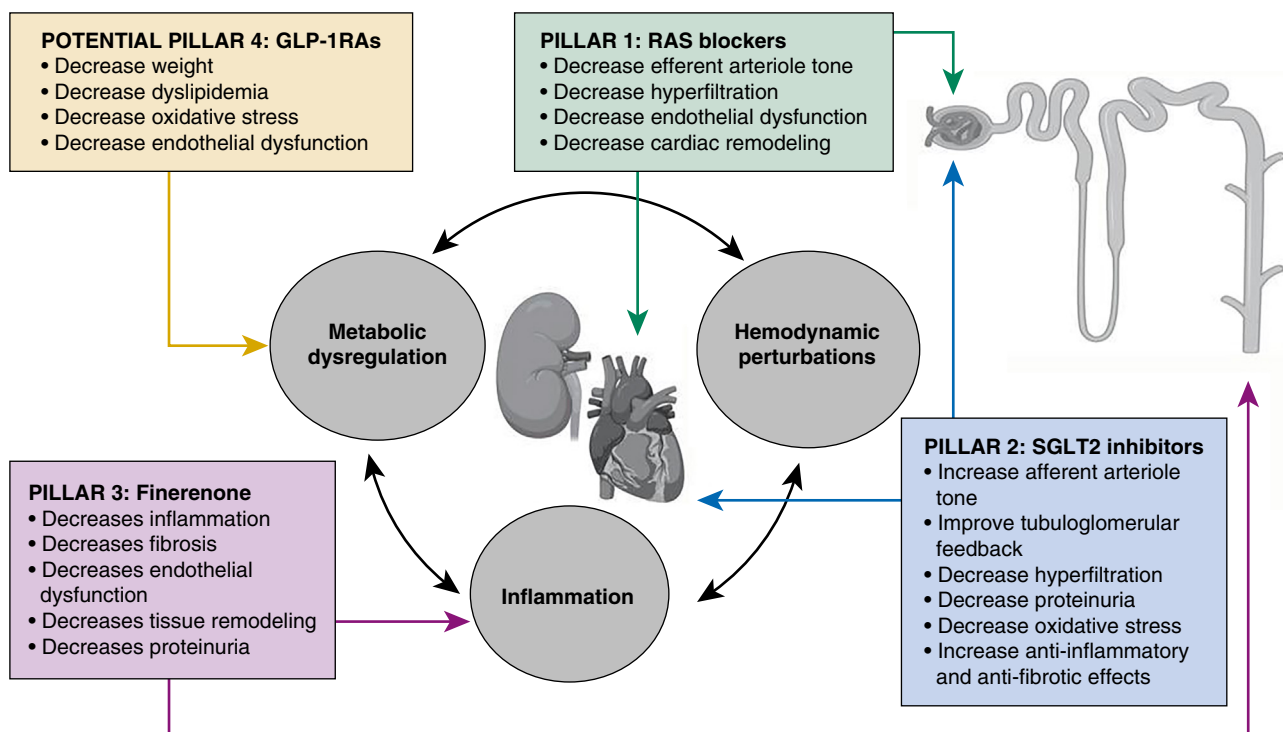


Fig. 54.12 The pillars of treatment for diabetic kidney disease to slow the progression of chronic kidney disease. GLP-1 RA, glucagon-like peptide-1 receptor agonist; RAS, renin-angiotensin system; SGLT-2, sodium-glucose cotransporter 2. (From Naaman, S. C. and G. L. Bakris. "Diabetic Nephropathy: Update on Pillars of Therapy Slowing Progression." *Diabetes Care* 2023 46(9): 1574-1586.)

benefits, without any evidence of dilution of effects.^{243,244} Finally, a modeling study combining data from multiple trials used actuarial methods to estimate the lifetime benefits of combination therapy with SGLT2 inhibitors, GLP1RAs, and nsMRAs in persons with type 2 diabetes and albuminuria (UACR >30 mg/g) reported a hazard ratio of 0.65 (95% CI, 0.55–0.76) for major cardiovascular events (MACEs). Considering a 50-year-old patient commencing therapy, combination treatment was associated with a predicted gain in mean event-free survival of 3.2 years for MACE, 3.2 years for hospitalization with heart failure, 5.5 years for CKD progression, 2.2 years for cardiovascular death, and 2.4 years for all-cause death.²⁴⁵ Once completed, the ongoing CONFIDENCE (NCT05254002) RCT will also elucidate the impact of empagliflozin alone, finerenone alone, or the combination of empagliflozin and finerenone, on change in UACR.²⁴⁶ Nonetheless, the absence of direct evidence on the effect of combination treatments with multiple classes of medications should not dissuade clinicians from ensuring all eligible people with CKD are started on available kidney-protective medications in line with existing clinical guidelines (see [Tables 54.2 and 54.4](#)). Indeed, some authors have suggested that we should adopt an approach of rapid initiation or escalation of the “four pillars” of kidney-protective therapy in persons with type 2 diabetes and the highest risk of adverse outcomes.²⁴⁷

INTERVENTIONS TO REDUCE CARDIOVASCULAR RISK ASSOCIATED WITH CHRONIC KIDNEY DISEASE

Following a seminal report in the early 1970s describing accelerated atherosclerosis in persons receiving maintenance hemodialysis,²⁴⁸ a large body of research has focused on the relationship between CKD and cardiovascular disease. It is now clear that increased cardiovascular mortality occurs early in the course of CKD (e.g., in the setting of microalbuminuria in persons with diabetes²⁴⁹ and when the GFR starts to decline²⁵⁰ in stage 3 CKD). Furthermore, the rate of CVEs associated with CKD is substantial, so reduced eGFR is now regarded as an effective marker (or “risk equivalent”) of premature cardiovascular disease, similar to diabetes. For this reason, any strategy for the management of CKD should also include interventions to reduce the associated risk of cardiovascular disease.

The pathogenesis and management of cardiovascular disease associated with CKD are discussed in detail in [Chapter 42](#) and are therefore reviewed only briefly here. As discussed throughout the text earlier, the interventions recommended for optimal kidney protection also contribute to reducing the risk of adverse CVEs. Thus lifestyle measures including smoking cessation, weight loss, and dietary sodium restriction afford both renal and cardiovascular protection. In addition, the treatment of hypertension with RAAS inhibitors and a lower blood pressure target of 130/80 mm Hg or SBP < 120 mm Hg if appropriate, central to achieving kidney protection, also reduces cardiovascular risk. SGLT2 inhibitors, MRAs, and GLP1RAs also confer cardioprotective benefits in people on maximal background RAAS inhibitor treatment, irrespective of baseline kidney

Table 54.4 Comprehensive Strategy and Therapeutic Goals for Achieving Maximal Kidney Protection in Persons With Chronic Kidney Disease

Intervention	Goals
ACE inhibitor or ARB treatment	UACR <30 mg/g (<3 mg/mmol) GFR decline <1 mL/min/year
Additional antihypertensive therapy	BP <130/80 mm Hg if urine albumin excretion >30 mg/day BP <140/90 mm Hg if urine albumin excretion <30 mg/day
Start SGLT2 inhibitors, nsMRA, and GLP1-RA in all eligible patients	UACR <30 mg/g (<3 mg/mmol) GFR decline <1 mL/min/year
Weight loss if patient is overweight	Aim for 5% weight loss
Dietary salt restriction	<5 g/day (equivalent to 90 mEq sodium/day)
Dietary protein restriction	Avoid high-protein intake >1.3 g/kg/day In stages 4 and 5 CKD, consider reducing intake to 0.8 g/kg/day
Tight glycemic control	HbA _{1c} <7.0% unless considered high risk for hypoglycemia
Smoking cessation	Complete cessation
Lipid-lowering therapy	Total cholesterol <200 mg/dL (5.2 mmol/L) LDL cholesterol <100 mg/dL (2.6 mmol/L)

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BP, blood pressure; CKD, chronic kidney disease; GFR, glomerular filtration rate; GLP1RA, glucagon-like peptide 1 receptor agonists; HbA_{1c}, hemoglobin A_{1c}; LDL, low-density lipoprotein; nsMRA, nonsteroidal Mineralocorticoid Receptor Antagonist; UACR, urine albumin-to-creatinine ratio.

function, UACR, or a diagnosis of diabetes.¹⁸⁵ Whereas SGLT2 inhibitors have a more pronounced effect on reducing the incidence of heart failure as compared with atherosclerotic CVEs, GLP1RA has a more pronounced effect on reducing atherosclerotic CVEs as compared with heart failure. Therefore the combined use of SGLT2 inhibitors and GLP1RA should be considered in people with CKD at high cardiovascular risk.

One exception to the principle that most interventions afford renal and cardiovascular protection is treatment with statins. The largest RCT to date found a 17% RR reduction in atherosclerotic CVEs in participants randomly assigned to statin (with or without ezetimibe) therapy²⁵¹ but no benefit with respect to CKD progression.²⁵² The 2024 KDIGO guideline for lipid management in CKD recommends treating all persons aged 50 years and older with CKD not treated with dialysis or renal transplant with a statin or statin and ezetimibe, regardless of low-density lipoprotein cholesterol levels. For persons with CKD younger than 50 years, the guideline recommends similar treatment for persons with a 10-year risk of CVE >10%, type 2 diabetes, prior ischemic stroke, or known coronary disease including myocardial infarction or coronary revascularization.⁸

Clinical Relevance

All persons with chronic kidney disease should be regarded as having a high risk of cardiovascular disease, and management should be offered to reduce this risk.

INTERVENTIONS TO MANAGE COMPLICATIONS OF CHRONIC KIDNEY DISEASE

ANEMIA

The anemia of CKD results from a combination of reduced kidney erythropoietin production (a presumed reflection of the reduction in functioning renal mass), shortened red blood cell survival, and functional iron deficiency. Anemia—defined as a hemoglobin count lower than 13 g/dL in men and lower than 12 g/dL in women⁹—can develop well before the onset of uremic symptoms. Among more than 15,000 participants in the Third National Health and Nutrition Examination Survey (NHANES III), the prevalence of anemia increased from 1% at an eGFR of 60 mL/min/1.73 m² to 9% at an eGFR of 30 mL/min/1.73 m² and from 33% to 67% at an eGFR of 15 mL/min/1.73 m².²⁵³

If left untreated, the anemia of CKD is associated with several adverse effects. These include deterioration in cardiac function, decreased cognition and mental acuity, and fatigue. In cross-sectional studies, associations between anemia and an increased risk of morbidity and mortality, caused principally by cardiac disease and stroke, have been described in persons on dialysis.²⁵⁴ In addition, anemia may influence the progression of CKD. In persons with inherited hemoglobinopathies, chronic anemia is associated with glomerular hyperfiltration that eventuates in proteinuria, hypertension, and ESKF.²⁵⁵ In several longitudinal studies, lower hemoglobin value was identified as a risk factor for CKD progression^{256,257} and ESKF.²⁵⁸ Whether this reflects more severe kidney disease in persons with lower hemoglobin values or more rapid progression secondary to low hemoglobin value and oxygen carriage per se is unclear.

Two small, randomized studies revealed kidney-protective benefits when anemia was corrected with erythropoietin.^{259,260} By contrast, enthusiasm for the normalization of hemoglobin has been tempered by the results of several large RCTs that revealed no benefit or adverse effects. In two studies in which effect on left ventricular mass was the primary endpoint, as well as the Trial to Reduce Cardiovascular Events with Aranesp Therapy (TREAT),²⁶¹ there was no effect of higher versus lower hemoglobin target on rate of decline in GFR.^{262,263} Moreover, in the Cardiovascular Risk Reduction by Early Anemia Treatment with Epoetin Beta (CREATE) study, achievement of a higher hemoglobin target (13–15 mg/dL) was associated with a shorter time to initiation of dialysis than was achievement of the lower target (10.5–11.5 mg/dL).²⁶⁴ Further concern was provoked by serious adverse effects associated with higher hemoglobin targets, including increased rate of mortality²⁶⁵ and increased risk of stroke.²⁶¹ Emerging treatments for anemia such as hypoxia

inducible factor prolyl hydroxylase inhibitors (HIF-PHIs) have been shown to be noninferior to erythropoietin for the management of anemia. The effect of HIF-PHIs on CKD progression remains unclear.

Current KDIGO recommendations are therefore to treat symptomatic anemia in CKD with erythropoietin or iron supplementation, or both, to partially correct the hemoglobin and achieve a range of 10 to 11.5 mg/dL.²⁶⁶ Hemoglobin value should not exceed 13 g/dL.²⁶⁶ The pathogenesis and treatment of anemia in CKD are discussed further in [Chapter 53](#).

MINERAL AND BONE DISORDER

Numerous cohort studies have shown strong associations between disorders of mineral metabolism and fractures, cardiovascular disease, and mortality (reviewed by the KDIGO CKD-MBD work group²⁶⁷ (see also Chapter 52). Some caveats are relevant here. Most of the studies reporting observational data and relationships between individual parameters and clinical outcomes have been performed in dialysis populations. The limited data regarding the prevalence of biochemical and hormonal abnormalities in stages 3 to 5 CKD often do not include analyses by primary disease, which can influence the natural history of mineral and bone disorders.

Changes in mineral metabolism and bone structure are detectable much earlier in CKD than had previously been considered. There is a slow decline in levels of 1,25-dihydroxyvitamin D and 25-hydroxyvitamin D, starting once the GFR is in the range of 60 to 70 mL/min. 1,25-Dihydroxyvitamin D values were correlated positively with GFR and negatively with the log of plasma parathyroid hormone and serum phosphorus concentrations. The plasma parathyroid hormone concentration rises later in the progression of CKD but more exponentially when GFR falls below 45 mL/min/1.73 m².²⁶⁸ Calcium and phosphorus values do not generally become abnormal until GFR falls below 40 mL/min/1.73 m², and this occurs more commonly below 20 mL/min/1.73 m².²⁶⁹

In the community-based Kidney Early Evaluation Program (KEEP) and NHANES cohorts, there is evidence that parathyroid hormone levels increased early in persons with stage 3 CKD, typically while calcium and phosphorus levels remained normal.^{269,270} These findings highlight the importance of monitoring parathyroid hormone along with calcium and phosphorus in individuals with eGFR <60 mL/min/1.73 m². Investigators have also described an earlier biomarker of altered mineral and bone disorder in persons with CKD. Fibroblast growth factor-23 (FGF-23) regulates phosphorus metabolism and is associated with mortality in persons on dialysis.²⁷¹ High levels of this growth factor, defined as being above 100 reference units/mL, were more common than secondary hyperparathyroidism and hyperphosphatemia at all levels of eGFR in a cohort of almost 4000 persons with stages 2 to 4 CKD.²⁷² Further analysis in the same cohort identified elevated FGF-23 levels as an independent risk factor for progression to ESKF and mortality.²⁷³ However, testing for this early marker is not routinely available, so we recommend monitoring according to the KDIGO guidelines.²⁷⁴

Thus monitoring of serum levels of calcium, phosphorus, and parathyroid hormone and of alkaline phosphatase activity should begin in stage 3 CKD. Few persons with mineral and bone disorder in CKD develop symptomatic disease until they have stage 5 CKD; therefore the main reason for monitoring is to implement early preventive treatment to suppress secondary hyperparathyroidism. In most persons with CKD who do not require dialysis, treatment of mineral and bone disorder is based initially on dietary phosphate restriction, followed by oral phosphate binders in order to maintain phosphorus in the normal range, although there is no evidence from RCTs that this affects clinical outcomes.

In persons with stages 3 to 5 CKD who are not on dialysis, the optimal level of parathyroid hormone is unknown. However, it is suggested that persons with levels of intact parathyroid hormone above the upper normal limit of the assay be first evaluated for hyperphosphatemia, hypocalcemia, and vitamin D deficiency.⁹ If parathyroid hormone level is progressively rising and remains persistently above the upper limit of normal for the assay despite correction of modifiable factors, treatment with calcitriol or vitamin D analogs (once phosphate is under control) may be considered. Treatment with the vitamin D analog paricalcitol has been reported to reduce albuminuria in persons with diabetic nephropathy already receiving treatment with an ACE inhibitor or an ARB,²⁷⁵ but further studies are needed to investigate the kidney-protective potential of vitamin D therapies. Detailed discussion of the pathogenesis and treatment of mineral and bone disorder in CKD are beyond the scope of this chapter and are presented in Chapter 52.

A STEPPED CARE APPROACH TO CHRONIC KIDNEY DISEASE

In addition to being useful for risk stratification, the KDIGO classification of CKD serves as a convenient framework to identify differing priorities for management at different stages. This “stepped care” approach is discussed in the following sections and summarized in Table 54.2.

STAGES 1 AND 2

At these stages of CKD, the diagnosis is based on the presence of albuminuria, hematuria, or structural kidney disease and an eGFR above 60 mL/min/1.73 m². Stage 1 disease is defined by a GFR >90 mL/min/1.73 m² (5.6% of the total U.S. population) and stage 2 by a GFR between 60 and 89 mL/min/1.73 m² (2.8% of the U.S. population).²⁷⁶

Persons with stages 1 and 2 disease do not have specific symptoms or complications of renal failure such as anemia or bone and mineral disorder. Those with symptoms may have a multisystem disease with secondary glomerular or interstitial disease (see Chapters 31–37). The majority of persons with stages 1 and 2 CKD are detected by routine or health care insurance-mandated screening and are visiting primary care or other physicians; in many countries, only a small proportion are ever evaluated by a nephrologist. With increased access to radiologic imaging, more persons are now identified with structural abnormalities such as polycystic kidney disease or single kidney.

The emphasis at these early stages should be on identification of specific renal diseases when present, appropriate referral to a nephrologist, and reduction of cardiovascular risk. A detailed family history is important because persons with a positive family history need more detailed investigation to allow early detection of inherited renal disease, particularly adult polycystic kidney disease. The following initial investigations are appropriate for assisting with risk assessment and informing decisions about referral to a nephrologist or urologist (see also Chapter 23):

1. Estimation of urinary albumin or protein excretion by measurement of ACR or PCR on a random urine sample. Persons with a urine protein measurement equivalent to 0.5 g/day (UACR, 300 mg/g or 30 mg/mmol) or greater should be referred for investigation by a nephrologist.⁹
2. Further urinalysis is needed to detect hematuria. For painless but visible hematuria, serious urologic causes—such as bladder, renal cell, and, less often, prostatic cancers—must be confirmed or ruled out, particularly in persons older than age 50 years, smokers, and those with a family history of renal tract malignancy. They should generally be assessed by a urologist or nephrologist who has experience in screening for these conditions. Painless microscopic hematuria (nonvisible hematuria) is much more likely to be caused by glomerular disease, but referral to a urologist may be necessary to confirm or rule out renal tract malignancy in persons at increased risk.²⁷⁷ Several national guidelines for the investigation of hematuria exist.^{277,278} Initial blood tests include measurements of creatinine, eGFR, urea, electrolytes (Na⁺, K⁺, HCO₃⁻, Cl⁻), bone and liver profiles, blood glucose, glycosylated hemoglobin, blood cell count, erythrocyte sedimentation rate, and (in men) prostate-specific antigen. Serologic screening for underlying myeloma, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, anti-glomerular basement membrane disease, and systemic lupus erythematosus may be indicated, depending on symptoms and the results of other investigations.
3. Abdominal ultrasonography to exclude structural abnormalities and determine the bipolar diameter of the kidneys is indicated if urinalysis results are abnormal if there is a strong family history of CKD, or if there is significant hypertension. Asymmetry regarding renal size may be suggestive of atherosclerotic renovascular disease, and angiography (computed tomography or magnetic resonance angiography) may therefore be helpful if this is suspected.

In general, persons with stages 1 and 2 CKD, who do not have a specific renal disease or significant proteinuria, require appropriate treatment and only annual monitoring of blood pressure, eGFR, and proteinuria. Those who develop an abrupt or sustained decline in GFR or worsening proteinuria should be referred to a nephrologist for further investigation and optimization of therapy.⁹

STAGE 3

In stage 3 CKD, GFR is between 30 and 59 mL/min/1.73 m². This is a significant stage because it represents the majority of persons in whom CKD is identified (stages 1 and 2 often remain undetected unless urinalysis is

performed) and because many of the complications start to manifest once the GFR drops below 45 mL/min/1.73 m². In addition, the rate of cardiovascular mortality increases substantially among persons with a GFR lower than 45 mL/min/1.73 m².^{250,279} Some national guidelines and the KDIGO guidelines have therefore split this stage into two: 3A, defined by reduced GFR of 45 to 59 mL/min/1.73 m², and 3B, defined by reduced GFR of 30 to 44 mL/min/1.73 m².^{9,280,281} Depending on the underlying etiology, stage 3 CKD may be appropriately managed by a collaboration of primary care with a nephrology service. One study that employed mathematic modeling techniques reported that multidisciplinary care for CKD is a cost-effective model that could help to reduce the need for dialysis and prolong life expectancy.²⁸² The aims of management at this stage are to identify specific renal disease, correct reversible causes of renal dysfunction, prevent or slow the progression of CKD, reduce cardiovascular risk, and treat the complications of CKD (usually in stage 3B CKD). Referral criteria are the same as for earlier stages of CKD, with the addition of anemia and mineral and bone disorder, which may necessitate specialist treatment. Monitoring of blood pressure, eGFR, and serum biochemistry profile, as well as complete blood cell count and evaluation for proteinuria, should be performed every 3 to 12 months, depending on risk profile and clinical circumstances (Table 54.5).

STAGE 4

Persons with stage 4 CKD have a high cumulative risk of cardiovascular death and progression to ESKF. Almost 66% of such persons experience either a renal event or a CVE over 5 years after diagnosis. In a population-based study,²⁸³ the proportions of persons who needed RRT during a 5-year observation period were 1.1%, 1.3%, and 19.9%, respectively, for stages 2, 3, and 4, and the respective mortality rates were 19.5%, 24.3%, and 45.7%. Not surprisingly, persons with stage 4 disease often make up a large proportion of those attending outpatient nephrology clinics. Achieving kidney protection remains an important goal to delay the onset of RRT for as long as possible, as does minimizing cardiovascular

risk. Blood pressure, eGFR, and serum biochemistry profile, including level of parathyroid hormone and complete blood cell count, should be monitored every 3 to 6 months.

As the GFR declines to below 20 mL/min/1.73 m², the focus should change to treating the complications of CKD and planning for RRT.²⁸⁴ Effective preparation for RRT requires input from multiple staff disciplines (medical, nursing, pharmacy, dietetics, psychology, and social work) and is best delivered in a multidisciplinary clinic. There is emerging evidence that persons prefer this approach to preparation and that such clinics are associated with better outcomes, at least in observational studies.^{285,286}

It is clear that late referral (<3 months before initiation of dialysis) for dialysis preparation is associated with significantly higher rates of mortality²⁵ and lower quality of life.²⁸⁷ Results of a Canadian study also indicated that even when referral was appropriately timed, there was a 53% higher rate of reaching the composite endpoint of death, need for transfusion, or subsequent hospitalization in those without an optimal start to RRT.²⁸⁸

The need for timely preparation for dialysis is clear and is emphasized in the majority of national guidelines. However, not all persons with stage 4 CKD progress to stage 5, and unnecessary preparation may do harm. Persons with stage 4 CKD should therefore undergo a formal assessment of their risk of progression to ESKF (see Chapter 19). This is particularly relevant in older persons, in whom the competing risk of death is an important consideration. A comprehensive approach to assessing the risk of ESKF versus death to assist decision making about RRT has been proposed by the European Renal Best Practice (ERBP) group.²⁸⁹ Preparation for initiation of dialysis requires multiple interventions to deal with both medical and psychosocial aspects. Persons require adequate counseling to assist them in the choice of dialysis modality and in coping with the psychosocial effects of starting dialysis. Elderly persons are often more accepting of dialysis than are younger persons, who may still be working or have family commitments. One large U.S. study confirmed that social support is important for persons on hemodialysis and peritoneal dialysis in terms of greater satisfaction, higher quality of life, and fewer hospitalizations.²⁹⁰ Timely formation

Table 54.5 Recommended Frequency of Monitoring by Stage of Chronic Kidney Disease

Variable	Stage 1 and 2	Stage 3	Stage 4	Stage 5
GFR and electrolytes	Every 12 months	Every 3-12 months	Every 3-6 months	Every 1-3 months
Proteinuria with ACR or PCR testing	Every 12 months	Every 3-12 months	Every 3-6 months	Every 3-6 months
Blood pressure	Each visit	Each visit	Each visit	Each visit
Calcium and phosphate levels	Every 12 months	Every 12 months	Every 3-6 months	Every 3 months
Parathyroid hormone level	—	Every 12 months	Every 3-6 months	Every 3-6 months ^a
Hemoglobin	Every 12 months	Every 12 months	Every 3-6 months	Every 1-3 months ^a

^aMonitoring of parathyroid hormone and anemia should depend on the previous results and specific treatment, if any, for these conditions.

Stable values with no specific treatment require less monitoring as indicated.

Monitoring should be individualized according to previous rate of GFR decline, risk assessment of future GFR decline (particularly high if heavy proteinuria, >1 g/day or equivalent), and current drug therapy.

ACR, Albumin-to-creatinine ratio; GFR, glomerular filtration rate; PCR, protein-to-creatinine ratio.

of vascular access, ideally a forearm arteriovenous fistula, is important to allow adequate maturation, modification if necessary, and repeat surgery in case of primary site failure. Peritoneal catheter insertion requires less maturation time but should be performed early enough to allow time for adequate training for peritoneal dialysis. See [Chapters 61–63, 69, and 70](#) for further discussion of preparation for dialysis and vascular access.

HEPATITIS B VACCINATION

Persons on hemodialysis have a small but significantly increased risk of exposure to hepatitis B and other blood-borne viruses. Severe outbreaks of hepatitis B in hemodialysis units have resulted in considerable morbidity and even mortality among susceptible persons and staff. Therefore persons with CKD in whom dialysis is anticipated should be screened for hepatitis B and C, as well as human immunodeficiency virus infection. Persons who are seronegative for hepatitis B surface antigen and hepatitis B surface antibody should be immunized and their antibody levels measured after vaccination. Because seroconversion rates decrease with GFR,²⁹¹ immunization should ideally occur in stage 3 in persons with a high risk of progression; however, in view of the large number of persons and lack of precision in predicting outcomes, it is usually delayed until stage 4. Seroconversion rates are low once dialysis has commenced, particularly in older persons.²⁹² Results of a meta-analysis of 12 studies indicated that increased seroconversion rates can be achieved by administering multiple doses of vaccine and preferably by the intradermal route.²⁹³ The mechanism is unclear, but this finding is consistent with the observed increased immunogenicity of other vaccinations after intradermal administration in persons with CKD and the rare cases of pure red blood cell aplasia after subcutaneous administration of recombinant erythropoietin.^{294,295}

PREEMPTIVE KIDNEY TRANSPLANTATION

Assessment and preparation for possible kidney transplantation should be undertaken before initiation of dialysis. The increase in death rates among waitlisted persons in comparison with transplant recipients is consistent, although still debated in view of methodologic issues, such as lead-time bias and unmeasured differences confounding these analyses.²⁹⁶ The clear benefits to recipients and the potential reduced demand for dialysis make a strong case for strategies to increase preemptive transplantation. Many countries and centers now permit kidney transplantation when the recipient's GFR is <15 to 20 mL/min/1.73 m² and if kidney function has decreased progressively over the previous 6 to 12 months.²⁹⁷ The optimal timing of preemptive kidney transplantation remains unclear. Neither higher nor lower pretransplantation GFR appears to be associated with superior allograft survival.²⁹⁸

STAGE 5

Once GFR declines to below 15 mL/min/1.73 m², priorities include maintaining optimal health and function, as well as achieving a planned and uncomplicated initiation of RRT. If persons have been referred in a timely manner,

preparation for RRT should already be complete, but ongoing psychosocial support is often necessary as persons come to terms with the imminent need to start therapy. The optimal time to initiate RRT remains a topic of debate.²⁹⁹ In a landmark RCT, persons with stage 5 CKD were randomly assigned to “early” (GFR = 10–14 mL/min/1.73 m²) or “late” (GFR = 5–7 mL/min/1.73 m²) initiation of dialysis. After a median of 3.59 years of follow-up, there was no difference in rate of survival or adverse events (CVEs, infections, or complications of dialysis) between the groups.³⁰⁰

We therefore recommend that the initiation of RRT should be individualized but in general should occur when the GFR falls below 10 mL/min/1.73 m² but before significant uremic symptoms or malnutrition occurs. To facilitate this timing, the frequency of monitoring of GFR, serum biochemistry, and hemoglobin, together with clinical assessment, should increase to every 1 to 3 months. Persons who decline RRT should continue to be treated for complications of CKD to optimize their quality of life and, if necessary, be referred to a palliative care service to allow adequate planning of their care once they develop symptomatic uremia.

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QUESTIONS

- Which of the following statements regarding the treatment of hypertension in chronic kidney disease is true?
 - Randomized controlled trials have provided clear evidence to guide recommendations for blood pressure targets.
 - The treatment of hypertension is effective in slowing chronic kidney disease (CKD) progression.
 - Treatment with a direct renin inhibitor is more renoprotective than treatment with an angiotensin receptor blocker (ARB).
 - Combination treatment with an angiotensin-converting enzyme inhibitor and ARB is recommended.
 - Current guidelines all recommend the same blood pressure targets.

Answer: b

Rationale: Though several trials and a large meta-analysis confirmed that the treatment of hypertension is effective in slowing CKD progression, trials that have investigated specific lower blood pressure targets have generally produced equivocal results. As a result, current guidelines recommend slightly different blood pressure targets (see the “Antihypertensive Therapy” section).

- Treatment with angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) has not been shown to provide renoprotective benefit in which of the following categories of people with chronic kidney disease (CKD)?
 - Type 1 diabetes with albuminuria
 - Type 2 diabetes with albuminuria
 - Nondiabetic CKD with albuminuria
 - Diabetic and nondiabetic CKD without albuminuria
 - Nondiabetic CKD with proteinuria >3 g/day

Answer: d

Rationale: Evidence of kidney-protective benefit from treatment with ACE inhibitors or ARBs has been clearly established by randomized controlled trials in all forms of CKD with albuminuria but not in the absence of albuminuria (see the “Pharmacologic Inhibition of the Renin–Angiotensin–Aldosterone System” section).

- Using the 2012 Kidney Disease Improving Global Outcomes (KDIGO) chronic kidney disease (CKD) classification system, identify the correct category for a patient with an estimated glomerular filtration rate of 42 mL/min/ 1.73 m² and urine albumin-to-creatinine ratio of 400 mg/g.
 - G3a A3
 - G3a A2
 - G3b A1
 - G3b A2
 - G3b A3

Answer: e

Rationale: See the “Classification of CKD” section.

- Which of the following treatments in people with diabetes is recommended in the 2024 Kidney Disease Improving Global Outcomes (KDIGO) guidelines for the management of chronic kidney disease (CKD)?
 - Angiotensin-converting enzyme (ACE) inhibitor for patients with CKD and hypertension
 - GLP1 receptor agonists (GLP1RA) as the preferred second-line agent for people with type 2 diabetes without adequate glycemic control
 - SGLT2 inhibitor for all people with an eGFR ≥ 20 mL/min/ 1.73 m²
 - Nonsteroidal mineralocorticoid receptor antagonist (nsMRA) for people with type 2 diabetes and residual albuminuria despite renin-angiotensin system (RAS) inhibitors
 - All of the above

Answer: e

Rationale: Combination treatment with RAS inhibitors, SGLT2 inhibitors, GLP1RA, and nsMRA is recommended in the 2024 KDIGO guidelines for kidney protection in people with type 2 diabetes. The role of GLP1RA for kidney protection may evolve given the forthcoming results of the pivotal kidney outcome FLOW trial, which was stopped early for efficacy (see the “Strategy for Maximal Kidney Protection: Aiming for Remission of Chronic Kidney Disease” section).

- Which of the following statements regarding lipid-lowering therapy in chronic kidney disease (CKD) is true?
 - Statin treatment is contraindicated in people on dialysis.
 - Statin treatment is renoprotective.
 - Lipid-lowering therapy reduced atherosclerotic cardiovascular events in a large randomized controlled trial (RCT) of patients with CKD.
 - Statins are contraindicated in people with CKD stage 5.
 - Statins do not lower cholesterol in people with CKD.

Answer: c

Rationale: A large, randomized, controlled trial (SHARP) reported a 17% relative risk reduction in atherosclerotic cardiovascular events in persons with CKD randomized to treatment with simvastatin with or without ezetimibe but no benefit with respect to kidney outcomes. The analysis indicated no difference in effects between participants receiving dialysis versus not on dialysis, though other trials have reported a lack of benefit with statin treatment in persons receiving dialysis (see the “Interventions to Reduce Cardiovascular Risk Associated with Chronic Kidney Disease” section).